基于框架工程的智能体系统

Agent Systems with Harness Engineering

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摘要

构建有效的智能体（agent）系统长期以来都是人工智能领域的前沿课题。近来基于大语言模型（LLM）的智能体的出现标志着一次范式转变，然而，即便是能力极强的模型，若在缺乏充分系统支持的情况下部署，也会在长程（long-horizon）任务中表现出系统性失败，包括丢失中间目标、错误地调用工具，以及无法吸收环境反馈。这些失败并非源于模型训练上的不足，而是反映出 LLM 的单轮生成式接口与现实世界问题求解所固有的有状态、迭代特性之间的结构性失配。在本文中，我们将围绕模型的支撑性基础设施形式化为**框架（harness）**，并提出**框架工程（harness engineering）**，即对框架及其所支撑的模型进行联合优化。二者之间的关系本质上是*协同增效*的：精心设计的框架能够通过结构化记忆、错误恢复和自适应上下文管理，释放模型的潜在能力；而能力更强的模型又使框架得以实现那些较弱推理器无法支撑的复杂工作流。我们提出了一套结构化的框架组件分类体系，涵盖智能体工作流、记忆系统、技能库和多智能体编排，并分析各组件如何贡献于系统级性能，从当前实践中提炼出设计原则与架构权衡。我们进一步从脚手架（scaffold）侧与模型侧两方面回顾了优化策略，并汇总了软件工程、深度研究、工具使用、计算机使用与科学发现等领域的评测基准。通过将关注点从智能体*能完成什么*转向支撑性基础设施*如何使其能够完成*，本文旨在为通过有原则的框架工程构建可靠、可扩展、可控的智能体系统提供一份实用参考。

关键词：框架工程，智能体系统

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代码仓库：https://github.com/RUCAIBox/awesome-agent-harness

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1 引言

构建有效的智能体系统长期以来都是人工智能领域的前沿课题 [1]，其历史可追溯到早期的符号式智能体 [2] 和基于规则的设计 [3]——它们在简化、结构化的环境中表现出色，却难以应对复杂性与可扩展性。数十年来，强化学习 [4] 与机器人智能体 [5] 的进展带来了自适应行为；但现实世界的部署仍受制于脆弱的决策能力与有限的泛化能力。

近来基于大语言模型（LLM）的智能体系统 [6] 的出现标志着一次范式转变：通过利用丰富的常识知识、自然语言交互与复杂推理，LLM 使智能体能够在开放式场景中进行规划、使用工具并自适应——极大地提升了它们在软件自动化 [7, 8]、科学发现 [9, 10] 与个人助理 [11, 12] 等现实应用中的实用性。鉴于这些卓越能力，早期的基于 LLM 的智能体研发将重心更多放在模型本身，而非智能体系统内部的支撑机制上。

然而，随着任务日益复杂，一个根本性挑战浮现出来：即便是能力极强的 LLM，若在缺乏充分系统支持的情况下部署，也会在长程任务中表现出系统性失败 [13, 14]。例如，模型常常在多步规划中丢失中间目标，以错误的参数或在不恰当的时机调用工具，并且无法吸收环境反馈以纠正先前的错误 [15]。这些缺陷并非仅仅是模型训练上的、未来通过扩大规模即可弥合的差距；它们反映出 LLM 的单轮生成式接口与现实世界问题求解所固有的有状态、迭代特性之间的结构性失配。由此逐渐形成一个共识：一个 AI 系统的实际能力，既取决于其底层模型的生成能力，也同样取决于其支撑性框架的设计。

因此，对支撑性系统的设计与开发予以同等重视同样重要。在 LLM 的语境下，智能体系统最初被定义为以 LLM 为基础、由记忆、工具、规划与动作等关键模块构成的整体性框架。此后该领域扩展到提示工程与上下文工程 [16, 17]，它们在不改变模型权重的前提下提升 LLM 的能力。更近期地，智能体框架通过重新设计智能体—环境接口本身而取得进展。与早期依赖自由文本动作或有限工具调用的系统不同，OpenClaw [11]、Hermes Agent [12] 等现代框架引入了能力强大的动作空间（例如 bash 命令、文件编辑、搜索操作）、有状态的沙箱执行环境以及对计算机的直接控制——旨在实现无需人工监督的自主、长程任务执行。这些设计选择将瓶颈从原始的 LLM 推理转移到稳健的系统脚手架上，使智能体从脆弱的概念验证转变为可面向现实任务部署的系统。

这一认识——即实际能力既取决于系统脚手架，也同样取决于模型本身——自然引出了对这一"支撑性系统"究竟包含什么的更精确刻画。在近期文献中，**框架（harness）**一词被专门用来指代围绕并扩展基础模型的基础设施 [13, 14]。我们据此将一个智能体系统分解为两个基本组成部分：一个作为推理引擎、负责语言理解、生成与决策的**基础模型**，以及一个掌管执行流程、工具调用、记忆管理、上下文构建、安全约束以及与外部环境多步协调的**支撑性框架** [18, 19, 20, 21]。没有框架，模型仍只是一个被动的文本生成器；只有当它被嵌入到一个经过恰当工程设计的框架中时，它才成为一个真正可用的智能体。

至关重要的是，这两个组成部分之间的关系不仅是组合性的，更本质上是*协同增效*的：对其中一个组件的改进会释放出另一个组件的潜在能力，从而产生任何一方单独都无法实现的系统级增益。例如，一个能力更强的模型能够遵循更细致的指令，从而使框架得以实现那些较弱推理器无法支撑的复

杂多步工作流。反过来，一个提供结构化记忆检索、错误恢复机制与自适应上下文管理的精心设计的框架，能够让一个能力中等的模型在那些本会超出其独立能力的长程任务上持续保持连贯的表现。框架工程沿着两条互补的战线推进这种协同：在脚手架侧，它设计更有效的工作流、记忆架构、技能库与编排机制，以构造并引导模型行为；在模型侧，它通过上下文工程与智能体训练对基础模型本身进行适配，以更有效地发挥框架所赋予的能力。

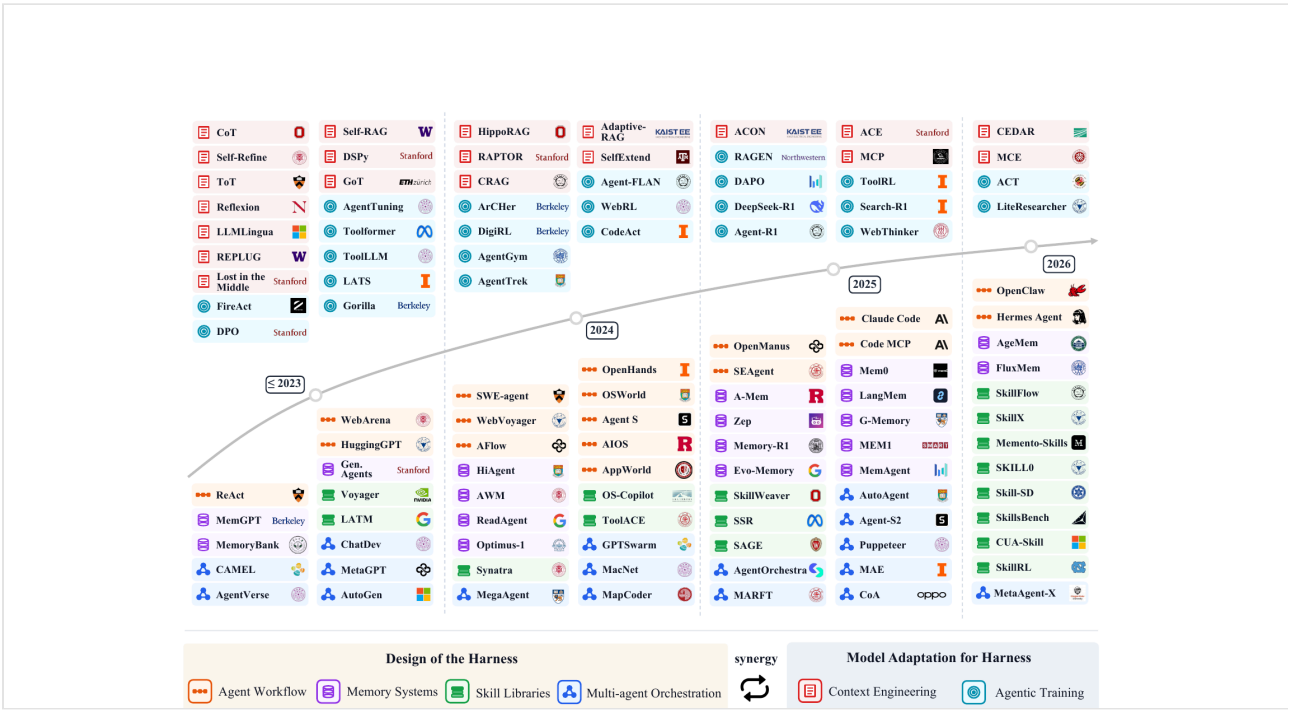


图 1： 具备框架工程的代表性智能体系统时间线。下半部分呈现脚手架侧的优化，涵盖四类：**智能体工作流、记忆系统、技能库**与**多智能体编排**。上半部分呈现模型侧的适配，包含两类：**上下文工程**与**智能体训练**。这两个维度本质上是协同增效的——脚手架设计塑造了为模型适配提供信息的交互模式，而模型能力的提升又反过来使更复杂的脚手架架构成为可能——二者共同推动更强大、更稳健、更具泛化能力的智能体系统的发展。

我们据此将**框架工程**形式化为对上述两个组件的联合优化：在设计框架以适配模型的优势与局限的同时，也对模型进行适配，以便利用框架所提供的能力支撑（affordance）。正是这种双向的协同优化，赋予了智能体系统应对复杂现实任务所需的可靠性、适应性与可控性。

在本文中，我们着重考察支撑性基础设施如何使有效的智能体行为成为可能，将分析的焦点从智能体的能力转向产生这些能力的机制。具体而言，本文的贡献有三方面：

- 我们将框架工程形式化为一种独立的研究范式，确立其概念基础并厘清它与相邻范式的关系。这为理解那些将语言模型提升为自主智能体的基础设施提供了统一的术语体系。
- 我们提出了一套结构化的框架组件分类体系，涵盖智能体工作流、记忆系统、技能库与多智能体编排，并系统地分析各组件如何贡献于系统级性能，从当前实践中提炼出设计原则、架构权衡与经验性最佳实践。
- 我们全面回顾了同时作用于脚手架与底层模型的优化策略，并汇总了在深度研究、软件工程、工具使用、计算机使用与科学发现等领域评估框架有效性的评测基准。

本文余下部分组织如下。第 2 节确立框架工程的概念基础。第 3 节详述框架设计的主要组件。第 4 节考察模型适配技术，包括上下文工程与智能体训练。第 5 节汇总深度研究、软件工程、工具使用、计算机使用与科学研究等领域的评测基准。第 6 节指出尚待解决的挑战与未来方向。

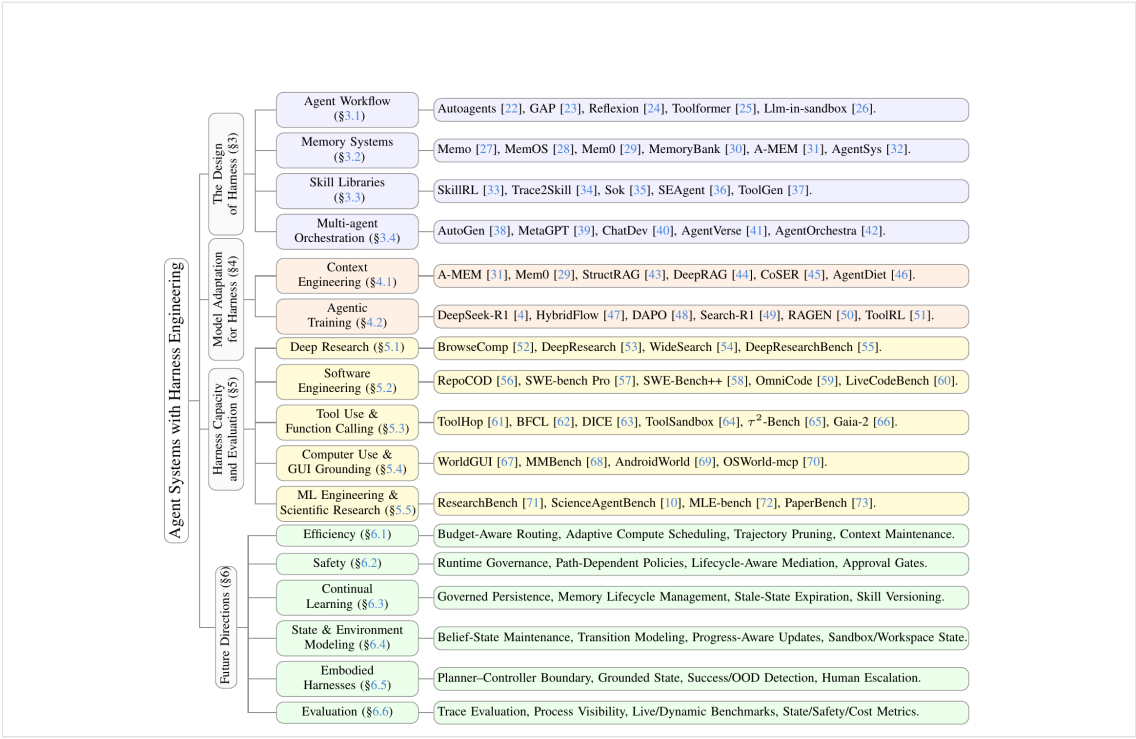


图 2：本文结构总览。

2 框架工程的基础

一个智能体系统的实际能力不仅取决于基础模型，也同样取决于支撑它的编排层（orchestration layer）[74, 75]。这一层组织推理步骤、居中处理工具交互，并在跨越较长任务时域的过程中掌管迭代执行 [76, 77]。我们将对其的系统性研究与设计称为**框架工程**，并特别关注：当智能体系统从简单的模型−环境耦合，演进为能力越来越强、越来越持久的辅助形式时，其范畴是如何扩展的。在本节中，我们通过三种逐步拓宽的视角引入这一概念（第 2.1 节），追溯其跨越三个演进阶段的发展（第 2.2 节），并厘清它与相关概念的关系（第 2.3 节）。

2.1 框架工程的概念

框架工程这一概念可以有多种解读。我们将它们组织为三种逐步拓宽的视角，每一种都涵盖了前一种。

视角一：作为结构性分解的框架工程。第一种、也是最基本的视角提供了一种结构性分解：一个智能体系统由两个构成性要素组成，即一个基础模型及其支撑性框架。基础模型提供语言理解、推理与生成方面的核心能力，而框架则构成了在模型与外部环境之间居中处理一切交互的编排层。具体而言，框架掌管工具调用（例如 API 调用与代码执行）、上下文维护（例如对话历史与记忆管理）、状态持久化（例如中间结果的存储）以及执行控制（例如终止条件与重试策略）。因此，系统级性能由两个组件共同决定：没有框架的基础模型仍是一个能够产生输出、却无法进行持续的、目标导向行为的推理引擎；只有当它与一个恰当设计的框架相结合时，才成为一个能够在现实世界环境中执行任务的、功能完备的智能体。然而，这一视角主要仍是描述性的，因为它指出了围绕模型的是什么，却未规定支撑性结构应当如何被设计或改进。

视角二：作为框架构建的框架工程。第二种视角从描述转向实现，将框架工程等同于智能体框架的设计与构建。从这一视角看，构建一个框架远不止于用一层薄薄的接口包裹一个模型；它意味着创建一套完整的、可复用的软件基础设施，以支持多步交互、工具集成与复杂的智能体工作流。在这一解读下，框架工程的目标与智能体框架开发的目标相一致：为开发者提供可扩展、稳健、开箱即部署的基础设施，用于构建智能体应用。尽管这一视角抓住了大量实际的工程工作，但它只聚焦于单一方向的优化，并未触及一个互补的问题：模型本身如何被适配以更好地利用支撑性基础设施。

视角三：作为协同增效的框架工程。本文采取一种更宽广、更整合的观点。我们不把框架工程视作结构性分解（视角一）或框架构建（视角二），而是将其定位为一种面向智能体系统的协同优化方法论。视角一聚焦于"什么构成了一个智能体系统？"，视角二考察"如何围绕模型构建脚手架？"，而视角三探究"如何对脚手架与模型进行联合优化，以最大化系统级性能？"这一协同性的表述涵盖两个互补的方向。第一个方向针对支撑模型的脚手架的优化（第 3 节），包括智能体工作流的构建、记忆系统的设计、技能库的策管（curation），以及多智能体协作的编排。第二个方向针对模型自身的优化，使其更好地与框架协作（第 4 节），涵盖上下文工程与智能体训练。任何一个方向单独都不够：一个架构精良的脚手架无法弥补一个缺乏基本推理能力的模型，一个能力极强的模型也无法在一个结构糟糕的系

统中发挥其全部潜力。正是通过对两侧的协同优化，智能体系统才能获得应对现实场景中复杂、长程任务所需的可靠性、可扩展性与可控性。

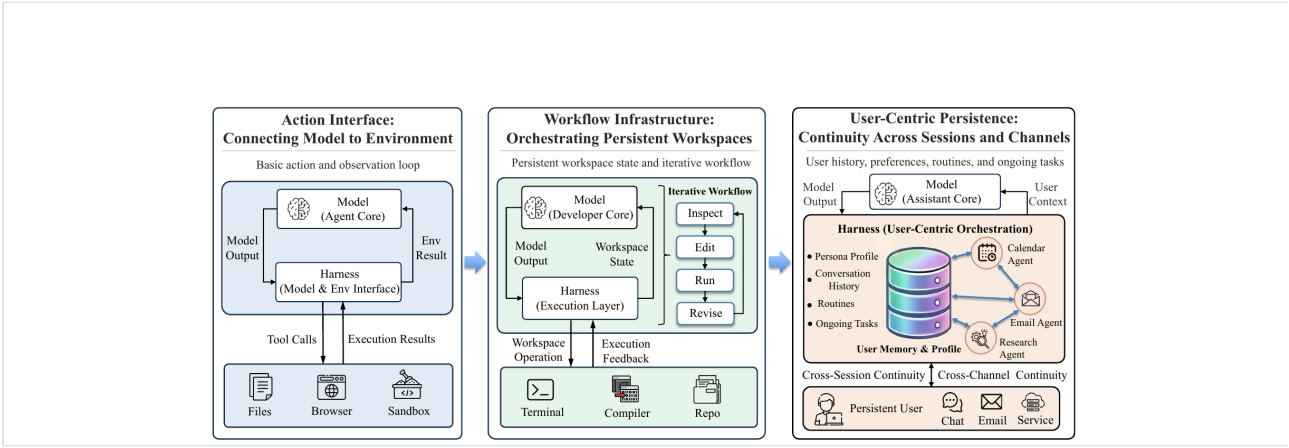


图 3：框架架构演进中的三个层次，展示了其范畴与职责的逐步扩展。**第 1 层（动作接口：连接模型与环境）：**框架将模型输出翻译为可执行的工具调用（例如文件操作、浏览器交互、沙箱执行）并返回环境观测，建立起基本的动作—观测循环。**第 2 层（工作流基础设施：编排持久化工作空间）：**框架维护持久化的工作空间状态，并在终端、编译器、版本控制系统等技术环境中编排"检查代码、应用编辑、运行命令、依据执行反馈修订"的迭代工作流。**第 3 层（以用户为中心的持久化：跨会话与跨渠道的连续性）：**框架管理用户画像、记忆、对话历史、惯例与进行中的任务，协调多个专用智能体，并提供跨会话与跨渠道的连续性。

2.2 框架工程的演进

随着智能体系统被部署到日益复杂的环境中，框架不得不支持更广泛的动作、更长时域的状态，以及更强大的委派控制形式。为满足这些不断增长的需求，框架工程经历了三个清晰的阶段，每个阶段都扩展了编排层所负责的范畴。在第一阶段，框架主要充当模型与环境之间的接口。在第二阶段，它成为在持久化技术工作空间中运行的开发者助理的执行层。在第三阶段，它充当个人助理的中枢，随时间推移并跨越通信渠道与个体用户保持连续性。

2.2.1 动作接口：连接模型与环境

框架的基础层建立起模型与其外部环境之间的连接。在这一层，框架将模型与文件系统 [78]、浏览器 [79] 与可执行运行时 [80] 等外部工具相连，把输出翻译为可执行动作，并将环境结果回送给下一步推理。这种闭环结构将一个基础模型从被动的文本生成器提升为一个能够作用于其环境的智能体 [81]。

其核心机制是一个"推理—动作—观测"不断重复的循环：模型生成一个动作，观测结果，并基于更新后的环境状态调整下一步 [82, 80]。一个代表性的例子是 ReAct [82]，它通过将推理轨迹与环境交互直接交织在一起，使这一循环显式化。后续研究沿着同一原则进行扩展，纳入了程序执行 [80]、API 调用 [25] 与多样的外部工具 [83]。更近期地，智能体系统将这些交互组织进统一的运行时中，使动作与观测能够在不同工具与环境间被一致地处理 [84, 78]。

2.2.2 工作流基础设施：编排持久化工作空间

在动作接口之上，第二层引入了持久化工作空间管理与多步工作流编排。这一层的框架不再仅仅在模型与单个工具之间架桥；它管理工作空间状态、协调一系列开发动作，并跨步骤携带执行上下

文，使模型能够检查代码 [7]、应用编辑 [85]、运行命令 [86]，并利用结果指导后续动作 [8]。这一闭环的开发循环将模型从一次性的代码生成器提升为一个能够迭代地精炼其输出、直至任务完成的开发者助理 [87]。

这一层的核心机制是一个"生成—执行—修订"不断重复的循环：模型产生代码，框架在工作空间中执行它，观测诸如测试结果或错误信息之类的结果，并将这些反馈回送以进行下一轮迭代 [74, 88]。一个代表性的例子是 SWE-agent [7]，它表明强劲的任务性能不仅取决于代码生成质量，更关键地取决于框架如何组织导航、反馈与迭代纠错。OpenHands [74] 与 Claude Code [88] 等系统进一步扩展了这一原则，将模型置于真实的开发环境中，赋予其对 shell、文件系统与版本控制的完全访问权，支持持续的编辑与验证循环，而非孤立的一次性生成。在软件工程之外，LLM-in-Sandbox [26] 表明，将同一工作空间基础设施虚拟化为一个最小化的代码沙箱，能够在一般的非编码任务上带来一致的增益。

2.2.3 以用户为中心的持久化：跨会话与跨渠道的连续性

第三层通过引入与个体用户跨会话、跨任务、跨通信渠道的连续性，将框架扩展到单次会话的任务执行之外。前一层在单个任务内维护工作空间状态，而这一层则将用户历史、偏好、惯例与进行中的任务作为持久化状态来管理，从而实现基于先前交互的个性化辅助 [89, 90]。这一纵向的交互循环将模型从无状态的任务执行者提升为一个能够随时间持续进行情境感知互动的个人助理 [91, 92]。

这一层的定义性要求是：随时间推移与同一用户保持连续性，即便任务跨越不同的会话或通信渠道 [93, 90]。这使得更长期、个性化的辅助成为可能，能够累积关于用户需求的知识并相应地自适应。一个代表性的例子是 OpenClaw [94]，它将助理视作一个持久的、面向用户的智能体，支持记忆、画像管理，以及在真实工具和长程交互中的委派动作。

2.3 与相关概念的联系

框架工程是一个新兴概念，与智能体系统研究中若干已确立的观念相交叉。在本小节中，我们厘清框架工程如何与这些相关概念相联系并在其基础上构建。

智能体框架（Agent Framework）。与框架工程联系最紧密的概念是智能体框架。这类框架为构建、部署与管理智能体应用提供可复用的基础设施，通常涵盖工具接口、执行环境、交互协议，以及对多步任务完成的支持 [76, 84]。它们的核心贡献在于提供一个具体的实现基础，把智能体行为变成一个可运行的过程。尽管二者有共通之处，但它们处于不同的层次，回答的是本质上不同的问题。智能体框架聚焦于"如何实现一个智能体系统"这一工程问题——即通过具体的代码与软件架构来实现设计。相比之下，框架工程处理的是"如何系统地优化一个智能体系统的整体性能"这一方法论问题，它不仅涵盖对脚手架的改进，也涵盖对模型本身的适配。这些优化策略在不同框架与模型间是通用的，并不依赖于任何特定框架的实现细节。因此，一个智能体框架可被视作框架的一个具体实例，而框架工程则充当指导此类实例设计的元层次（meta-level）学科。在本文中，我们采取这一更宽广的观点，并系统地考察通过脚手架优化与模型优化这两个互补方向来构建更有效智能体系统的设计原则。

其他相关概念。除智能体框架外，还有另外三类概念与框架工程密切相关，每一类都贡献了一种被框架工程整合进统一的系统级方法论的关键要素。

- **提示工程与上下文工程**聚焦于设计或组织输入，以从 LLM 中引出更好的响应 [95, 96, 97]。框架工程将这些技术纳入其模型侧优化的一部分，同时进一步在整个智能体工作流中组织多步执行、状态管理与控制流。
- **工具使用**考察 LLM 是否以及如何调用外部函数或 API [98, 99]。框架工程将工具使用作为其脚手架设计中的一个组件纳入进来，并额外处理何时调用工具、如何处理失败以及如何将观测整合进后续推理。
- **开发者助理与个人助理**按应用角色对智能体系统进行分类（例如编码支持或日常任务辅助 [74, 89]）。框架工程提供底层的编排结构，包括上下文管理、进度跟踪与稳健执行，使此类助理角色得以有效运转。

3 框架的设计

在第 2 节确立了框架工程的概念框架之后，我们现在转向脚手架侧：那些包裹基础模型并塑造其智能体行为的架构组件。这些组件并非作为孤立的模块运作，而是构成一条整合的流水线。智能体工作流在一个闭环中编排感知、规划与动作；记忆系统将智能体的有效时域扩展到单个上下文窗口之外；技能库累积可复用的行为单元；多智能体编排则将系统扩展到超出任何单个模型实例容量的任务。我们将逐一考察各组件，重点关注那些在已部署系统中被证明有效的设计原则。

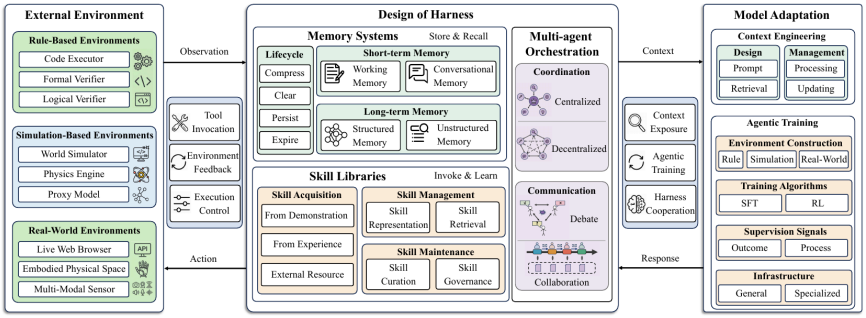


图 4： 框架设计空间及其与外部环境和模型适配交互的总览。**左：**外部环境为智能体执行与学习提供锚定（grounding）信号。**中：**共同构成框架基础设施的四个主要脚手架侧组件。智能体工作流掌管"感知—规划—动作"执行循环；记忆系统跨步骤、跨会话维护持久化上下文；技能库累积可复用能力以避免冗余探索；多智能体编排将复杂任务分解到各专用智能体上。**右：**通过上下文工程与智能体训练进行的模型侧适配，通过同时塑造底层模型的输入信息与推理行为来补充脚手架。脚手架侧的构建与模型侧的适配共同构成框架工程的两大互补支柱，二者的协同优化共同塑造了整体的智能体能力。

3.1 智能体工作流

智能体工作流是智能体框架的运行核心，它将一个无状态的语言模型转变为一个目标导向的系统 [100, 101]。由于语言模型执行单次前向传递后即终止，那些看似持续的行为完全是由框架构建出来的——框架维护消息历史，并在每次调用时重放它 [102]。框架通过将模型包裹在一个循环中来弥补这种无状态性：调用模型、解析响应、执行所指示的工具，并将结果回送以进行下一轮迭代。这一循环引出了三个紧密耦合的阶段。框架首先感知环境以收集任务相关信息（第 3.1.1 节），随后基于累积的状态规划下一步（第 3.1.2 节），最后执行所选动作，其结果成为下一轮迭代的输入（第 3.1.3 节）。

3.1.1 环境感知

环境感知是每轮循环迭代的入口。在智能体能够规划或行动之前，框架必须构建一个对当前环境的表征，它既要足够准确，又要足够紧凑以适配上下文预算（context budget）。这涉及两个互补的挑战：从原始环境中提取任务相关信息，并将其编码为下游规划能够可靠消费的形式。

观测提取（Observation Extraction）。核心挑战在于从复杂界面中提取空间与逻辑关系，无论界面是网页 [103, 104]、桌面应用 [105, 106] 还是移动应用 [69]。早期方法从 UI 层级结构中提取结构化特征并将其转换为文本表征 [107, 108]。这些表征易于解析，但往往庞大且冗余，从而催生了压缩技

术，在保留交互元素的同时剪除任务无关内容 [109]。纯文本表征缺乏视觉锚定，当布局偏离预期层级时会失效 [110]。基于截图的视觉提取解决了这一局限，并通过在大规模 GUI 语料上预训练 [111] 与在交互轨迹上进行强化学习 [112, 113] 进一步获得增益。然而，纯视觉方法仍受困于空间歧义。因此，当前的实践状态是通过多模态提取结合两个通道，在解析结构化元数据的同时结合截图，以获得元素级描述，包括控件类型、文本内容与空间属性 [114]。这一从文本到视觉再到多模态提取的演进并非单纯朝着更丰富输入的进军；每一步都引入了自身的代价。它反映出信息完整性、token 效率与提取可靠性之间的根本性权衡。每增加一个感知通道都能提供对环境更完整的描绘，但也消耗更多上下文预算，而每一个转换步骤都引入了产生噪声的机会。更丰富的感知并不总能带来更好的下游决策；工程上的挑战在于为给定任务最大化每个 token 所承载的有用信号。

状态表示（State Representation）。信息提取出来之后，框架还要把它编码成一种模型可以用来规划的表示形式。常见做法是采用 JSON [115]、DOM 风格的树 [7, 108] 或代码格式 [116] 等结构化标记，生成紧凑的文本描述。这类描述一旦过长，就会超出上下文窗口，或因信息过载而拖累规划质量 [117]。一些简单的启发式方法可以有效控制这种膨胀，例如屏蔽无关属性、在超出 token 预算时截断，或只保留最近的观测 [109, 118]。除了这种表层剪枝，还有一些方法借助一个辅助模型来摘要或过滤环境状态 [119]，但它们难以刻画各元素之间的关联。为了对这类关联建模，最新的研究把整个环境提炼为紧凑的、面向任务的世界模型，只保留与任务相关的对象、属性与状态转移 [120]；也有研究构建场景图 [121, 122] 这类显式结构，把环境压缩成一小组对规划最有用的实体及其空间配置。工业界把这一过滤思路推进得更远，并奉之为一条核心设计原则：感知接口应当被设计成"智能体—计算机接口"（Agent–Computer Interface），而非"人—计算机接口"（Human–Computer Interface）[123]。框架只在上下文窗口中呈现少数会实质影响决策的信号，而把冗长的细节写入文件、供智能体按需查询，从而把上下文预算花在决策价值最高的 token 上。

3.1.2 任务规划

任务规划决定了框架如何分解目标并在循环迭代间维护规划状态。复杂任务无法在单次推理步骤中求解，原因在于两个相互耦合的约束：自回归的误差累积导致一旦推理深度超过模型在单次传递中所能维持的程度，成功率便急剧下降；而有限的上下文窗口无法同时容纳目标、完整环境状态与所有相关约束。任务分解通过将一条长推理链切分为由校准点（calibration point）分隔的若干较短的链来同时缓解这两个约束，其中每条新链都从一个经工具执行确认的状态开始，而非从一个未经验证的中间 token 开始。更细的分解插入更多校准点并提升端到端的可靠性，但代价是额外的循环迭代与更高的总 token 消耗。现有研究从两个视角处理这一问题：如何将一个复杂目标拆分为中间子任务，以及如何将这些子任务转换为可执行的动作序列。

任务分解（Task Decomposition）。早期研究将中间推理组织为日益灵活的结构，从线性链 [95, 124] 发展到分支树 [125]，再到允许路径合并与复用的图 [126]。这些方法依赖于未针对具体任务自适应的预定义推理结构。更近期的方法 [127, 128] 通过让智能体在执行开始前分析任务并选择合适的推理策略，缓解了这种失配。两个新兴方向则彻底超越了静态结构。第一个是跨会话的增量分解 [129]。当一个任务跨越多个上下文窗口时，单次规划会失效，因为没有单个会话能容纳完整的问题状态。一种实用的解决方案是让初始智能体产出一份完整的功能需求文档，供后续智能体连同中间产物与版本控制历史一起用来恢复规划状态。每个新的上下文窗口引入一个新的智能体，它在继续前一会话中断

之处之前，先从这种外部化状态中重建对任务的理解。第二个是由外部验证器驱动的自适应分解[123]。对于难以自然并行化的任务，一个已知良好的预言机（oracle）可充当参照点。在一个已部署的例子中，多个智能体协作构建一个 C 编译器；框架将源文件子集分配给智能体构建的编译器，用 GCC 编译其余部分，并交叉比对输出以检测差异。这将一个本质上串行的任务转变为可并行化的子任务，其边界由运行时反馈动态确定，而非事先固定。

规划生成（Plan Generation）。在分解之外，当每一步纳入显式的工具使用规格 [130]，以及当线性序列被替换为支持独立子任务并行运行的依赖感知结构时 [23]，规划质量会提升。规划的表示格式也影响循环效率。以自然语言描述的规划可能消耗数百个 token，而以结构化工具名与参数表达的同一规划所需 token 则少得多。然而，过于激进的压缩会剥离每一步背后的意图与假设，在失败后需要重新规划时显著降低重试尝试的质量 [129]。这一张力表明，规划生成最好不要被当作一次性的输出，而应被当作一个由框架循环持续验证与调整的"活的产物"。

3.1.3 动作执行

一旦生成了规划，智能体必须将其翻译为影响外部环境的具体操作。我们从两个视角考察动作执行：智能体如何选择并调用离散的外部工具，以及它们如何在动态、有状态的环境中运作。

工具调用（Tool Invocation）。工具调用关注框架如何使智能体为给定任务选择并调用合适的工具。早期研究 [25] 将工具 token 直接嵌入模型词表，只支持顺序的单工具调用。后续方法 [131] 将智能体输出解析为可执行的函数调用，支持分支与多步组合。为使这些能力在已部署场景中落地，实用系统 [7, 132] 将文件系统、shell 命令等组件以带有显式模式（schema）的结构化工具接口暴露出来。一条趋于一致的研究脉络表明，可靠的工具调用必须通过可执行的环境反馈来学习，而非事先指定。那些利用运行时执行结果作为直接信号、通过强化学习训练智能体的方法，在单工具可靠性与多步组合两方面都显示出一致的增益 [133, 134, 135]。在调用机制之外，工具接口的粒度是一个关键的框架设计选择。诸如 bash 之类的通用接口提供最大灵活性，但几乎不给框架留下实施白名单、校验参数或隔离权限的能力。带有显式模式的专用接口，如模型上下文协议（Model Context Protocol，MCP）中定义的那些，以灵活性换取接口级的安全检查 [136]。类似地，SquRL [137] 等领域专用系统通过从目标领域抽象出面向任务的工具 [138] 来提供专用接口。恰当的平衡取决于部署领域的安全要求。一个相关的扩展性挑战出现在可用工具数量达到数百个时。在传统模式下，每个工具结果都返回上下文窗口，长工具链会累积可观的 token 开销。"用 MCP 执行代码"（Code Execution with MCP）[139] 通过让模型在沙箱中写出单个代码块、按序调用多个工具来解决此问题；只有最终结果返回上下文，从而将一条多轮工具链折叠为单次模型调用。

环境交互（Environment Interaction）。环境交互关注智能体如何在有状态环境中执行动作，这些动作的后果会跨步骤持续存在。早期研究 [82] 将智能体置于具有无界动作空间的原始环境中，使得难以可靠地将意图翻译为可执行动作。后续研究 [7] 引入了结构化的智能体–环境接口，暴露可用动作、预期格式与由此产生的状态变化。在此基础上，过滤方法 [140] 将当前规划与基于环境的奖励信号结合，在执行前丢弃不可行或不相关的动作。即便动作结构良好且已过滤，可靠的交互仍要求执行与宿主系统相隔离。沙箱机制 [79, 78] 将智能体行为限制在受控容器内，限制文件系统访问、限制网络暴露，并支持在多次尝试之间进行干净的状态重置。然而，仅靠隔离并不能保证安全；近期分析

[141] 表明，被沙箱化的智能体在规划、记忆与执行边界处仍表现出系统性脆弱性，需要主动的轨迹级审计才能检测。

> **讨论**
>
> - 有限的上下文窗口是感知、规划与执行所竞争的共享预算：丰富其中一个阶段必然使其他阶段"挨饿"。
> - 智能体循环的核心价值不在于逐步走完子任务，而在于创建校准点，让外部反馈纠正累积的模型漂移（drift）。
> - 循环的可靠性更多取决于框架设计；缺乏恰当支持时，智能体会把所有工作压缩进单个上下文窗口，并在接近上限处过早终止。

3.2 记忆系统

在即时执行循环之外，一个有能力的框架必须在超出任何单个上下文窗口容量的推理步骤间维持智能体的连贯性。这需要显式的记忆系统来掌管"保留什么、压缩什么、卸载什么"。其指导原则不是尽可能多地记住，而是只持久化那些对未来决策具有复用价值的信息 [142, 129]。我们沿两个维度考察记忆系统：短期记忆（第 3.2.1 节）管理单个会话内的信息，而长期记忆（第 3.2.2 节）累积超越即时上下文而持续存在的知识。

3.2.1 短期记忆

短期记忆掌管智能体如何在单个会话内跟踪并利用信息。框架必须主动决定在上下文窗口内保留、压缩或丢弃哪些中间状态。关键的设计洞见是区分可重建信息（如可被重新获取、因而可安全丢弃的工具输出）与不可替代信息（如必须保留或压缩的推理结论与用户决策）[142]。我们讨论两个互补的方面：管理活跃推理期间中间状态的工作记忆（working memory），以及跨轮次维护对话历史的对话记忆（conversational memory）。

工作记忆。工作记忆为任务执行期间产生的中间状态提供动态存储。早期设计依赖于将推理步骤、动作与观测交织在一起的线性轨迹 [82, 143]，但线性格式难以应对涉及分支或回溯的非线性推理，常导致信息重复 [144] 或关键上下文丢失 [27]。后续研究引入了结构化的替代方案：基于图的方法 [145] 将中间想法组织为支持分支与重组的相连节点，而基于工作流的方法 [146] 进一步通过可复用结构组织执行状态。在格式之外，工业实践揭示出并非所有中间状态都具有相同的生命周期。工具调用结果通常可被重新获取，应当首先清除；任务进度检查点则必须持久化以支持跨窗口的恢复 [142, 129]。这种生命周期感知的分解，将工作记忆管理从统一压缩推向基于可重建性的选择性保留。

对话记忆。在长对话中，来自早先轮次的重要信息（如用户偏好或先前决策）会随着历史增长超出上下文窗口而丢失。早期方法将对话历史存储在外部，并使用分层缓冲 [147] 或分页机制 [148] 按需检索相关内容，但它们以粗粒度运作，可能将相关上下文割裂开 [149]。后来的研究通过将历史划分为任务状态 [150]、用户画像 [151] 等功能单元，实现了更细粒度的切分，尽管仍孤立地处理每个片段。近期研究 [152, 153] 通过记忆元素间的显式链接来恢复跨片段依赖，而最新的努力 [154, 155] 则转向自适应管理，基于不断演化的对话状态学习保留、更新或丢弃什么。这种自适应方法已被大规模部署。例如，Claude Code 在长会话期间使用压实（compaction）来产生结构化摘要，在保留关键决

策的同时回收上下文预算 [142]，这表明可重建信息与不可替代信息之间的区分能够在生产系统中被付诸实施。

3.2.2 长期记忆

短期记忆管理活跃的上下文窗口，而长期记忆使智能体能够累积跨会话持续存在的知识。由于模型在两次调用之间不维护任何持久状态，所有用于存储、索引与检索长期知识的机制都必须由框架实现。从功能视角看，长期记忆分为三类，各有不同的生命周期 [156, 157]：语义记忆（事实、用户偏好、项目知识）、情景记忆（过往交互、失败与成功的轨迹）与程序记忆（任务规则与工作流）。每一类都需要不同的写入策略与检索机制。此外，并非所有长期记忆都应实时写入；必须立即生效的偏好属于热路径（hot-path）记忆，而经验提取与冲突消解则更宜在任务完成后异步处理 [157]。我们考察两种组织形式：以显式关系格式维护知识的结构化记忆，以及将信息存储为平铺文本的非结构化记忆。

结构化记忆。结构化记忆以关系形式组织累积的知识，如知识图谱 [158] 或显式链接的条目序列 [31]。早期基于图的设计 [159, 160] 将知识存为节点、将关系存为边，支持高效遍历，但通常只能容纳一种信息类型。后来的方法 [161, 162] 使用节点标签在单一结构中统一多种类型，但仍需要事先指定固定的模式，限制了对未见任务的泛化。更近期的工作 [31, 29] 移除了这种刚性，允许每个条目携带自身的上下文描述，并基于语义相似性与使用模式（而非预定义的类型）与其他条目链接。

非结构化记忆。现实世界的经验常以缺乏显式关系组织的平铺文本形式存在 [163]。早期研究 [148, 30] 存储自由格式文本并使用相似度匹配进行检索，但冗长的条目消耗大量空间，且过时信息难以更新。随后的一条研究脉络 [24, 164] 通过以自然语言保留过往交互与反思来使记忆显式化，便于检视、编辑并蒸馏为可复用的指南。然而，这些系统依赖手工设计的规则来决定存储或丢弃什么。近期研究 [154, 165, 166, 167, 168] 将记忆操作视作可学习的决策，而非固定的启发式规则，并扩展到长程个性化 [169] 与按需记忆扩展 [170]。

展望未来，长期记忆正从单个智能体的内部组件演变为一个共享的基础设施层。Mem0 [171] 与 OpenMemory [172] 等系统将记忆抽象为一个可移植的个人上下文层，通过 MCP 等协议暴露标准化接口，使不同的智能体能够按需访问同一记忆。在技术层面，记忆成为一项可复用的服务，而非每个框架各自独立实现的东西。在所有权层面，本地优先（local-first）的设计将数据存储在用户的机器上，使记忆跨越短暂的智能体会话而持续存在，并实现跨应用的上下文连续性。这一架构转变也模糊了记忆与技能之间的边界：当一个智能体反复从情景记忆中蒸馏可复用的过程并将其打包为可执行指令时，记忆便从情景形式过渡到程序形式，并最终成为一个可调用的技能 [173]。从这个意义上说，下一节讨论的技能库可被视作记忆系统的一种高级形式。

讨论

- 记忆的核心挑战不再是容量，而是治理：无界的累积会引入陈旧冲突、检索退化与持续的行为损坏。
- 上下文预算应被视作一项一等的运行时资源，需主动监控与守护，而非一个由模型不断填满直至溢出的被动容器。
- 智能体状态并非同质——对话历史、工具结果、任务检查点与长期知识在持久性与更新频率上各不相同，必须分别管理，而非混入一个不加区分的上下文中。
- 框架应施加一套有纪律的生命周期：压实与清除处理会话内的膨胀，而只有仍具价值的跨会话知识才进入持久层；可复用的过程进一步从情景记忆"毕业"为可调用的技能。
- 记忆正从内部模块演变为共享、可移植的基础设施层：智能体是短暂的，但记忆是长存的，借助用户自有的上下文实现跨应用的连续性 [171, 173]。

3.3 技能库

智能体工作流使行动成为可能，记忆系统保存状态，但二者都无法阻止智能体重新发现先前已解决问题的解法。技能库通过持久化可复用的行为单元来弥合这一缺口，使智能体能在类似情形再次出现时调用它们 [35, 174, 175]。从概念上看，技能库与第 3.2 节讨论的记忆系统处于一个连续谱上。当一个智能体反复从情景记忆中提取可执行的过程并将其编码为可执行指令时，记忆便逐渐从情景形式过渡到程序形式 [176]。在生产中，这一过渡越来越显式：技能被编写、版本化，并作为包含指令、脚本与资源的自包含包来分发 [177, 178]。因此，技能库设计的范畴不止于学习，还涵盖构建、运行时路由与生命周期治理 [179, 180]。本节考察构建与运营此类库时出现的三个核心问题：技能如何被获取（第 3.3.1 节）、它们在运行时如何被组织与检索（第 3.3.2 节），以及库在增长过程中如何被维护（第 3.3.3 节）。

3.3.1 技能获取

技能获取将成功的行为转化为可复用的能力。框架拦截成功的轨迹、触发提取例程，并将结果持久化以供日后检索。核心难点在于泛化：在一条轨迹中观察到的行为必须被充分抽象，以便迁移到原始上下文之外。现有方法从三个方向处理这一问题。

从示范中学习（Learning from Demonstration）。最直接的途径是从先前的成功行为中导出技能 [35, 174]。早期方法模仿工具调用日志 [25, 181, 182, 183] 或重放专家轨迹 [184, 185]，得到的技能能复现观察到的动作，但泛化性差 [186]。将完整轨迹注入上下文也会引入冗余 [187]。因此，后续研究将轨迹分解为可重组为更高层能力的原子单元 [34, 188]，或在先前记录中识别可复用结构作为更广泛技能创建的种子 [189]。更近期的方法进一步在组合过程中将文本知识作为结构化指导加以整合 [190, 191]。

从经验中学习（Learning from Experience）。智能体不必依赖外部示范，而可以通过自身与环境的交互来获取技能。早期的强化学习形式化 [192, 193] 产生迁移性差的任务特定策略 [194, 33]。后来的研究将问题重新表述为从交互中发现可复用的行为结构 [195, 196, 197, 198, 199, 200]：成功的动作序列被打包为超越原始情节而持续存在的技能。近期方法通过引导探索朝向环境中可能产生新颖、可泛化行为的区域，进一步提升了覆盖率 [201, 36, 202]。在一个极端情形下，Tool-R0 [203] 取消

了对任何预定义任务规格的需要，完全通过自博弈、以可执行环境反馈作为唯一学习信号来训练工具调用技能。

从外部资源中学习（Learning from External Resources）。技能根本不必源自智能体轨迹。工业实践越来越多地从文档、代码仓库与 API 定义中构建技能 [204, 177]。Anthropic 的 Agent Skills 框架 [177, 205] 将技能定义为基于目录的、包含指令、脚本与资源的包，智能体在运行时发现并加载它们。OpenAI 的 Skills API [178] 扩展了这一模型，提供可上传、管理并挂载到托管环境中的版本化捆绑包。在这些系统中，技能作为版本化的软件产物运作，具有类似于传统软件包的生命周期管理。

3.3.2 技能管理

一旦库中包含大量技能，就会出现两个组织性问题：如何表示技能以实现高效访问，以及如何在运行时选出正确的技能而不压垮模型的上下文 [206]。

技能表示。早期研究 [184, 35] 将技能编码为连续嵌入，支持基于相似度的索引，但牺牲了内部结构。后来的方法采用程序化格式，如可执行代码片段 [190]、结构化推理过程 [34, 195, 207] 或统一的工具使用模式 [208, 37]，使组合与迁移更显式 [194]。生产系统更进一步，将技能表示为包含指令、脚本、元数据与测试的完整包 [177, 178]。这种包级设计支持渐进式披露（progressive disclosure）：智能体最初只加载元数据或目录级摘要，并按需展开为完整指令 [177, 209]。由于任何技能在给定时刻只有一小部分占用上下文，库的规模可以远超全量注入方法所能支持的程度 [210, 179]。

技能检索。基于嵌入的相似度 [195, 184] 提供了一种自然的检索方法，但当任务需要组合多个技能时会退化，如多智能体协调 [211, 212] 或具身操作 [213] 中的情形。分层检索通过先解析粗粒度意图、再在更窄的子集中搜索来缓解这一问题 [208, 210]。情境敏感的检索将选择条件化于环境状态、先前轨迹或中间失败 [214, 215]，把检索变成一个迭代的决策过程 [216]。在生产规模下，库中包含数千个相互重叠的条目，检索成为一个可学习的路由问题。SkillRouter [179, 206] 通过一个在完整技能文本上运作的"检索–重排"流水线展示了这一点，表明只暴露名称与描述会降低路由准确率。这一发现表明技能内容本身是一个关键的路由信号，且专门的可学习路由器优于仅靠元数据的匹配。

3.3.3 技能维护

随着库的增长，挑战从获取与检索转向长期的健康。一个静态的库不可避免地会累积冗余、过时或冲突的条目，从而降低检索精度与执行可靠性。

库的策管（Library Curation）。随着工具 API 变化与外部依赖演化，技能在真实环境中会衰退 [217, 218]。框架必须检测执行失败、将其追溯到具体技能，并触发有针对性的更新。在库的层面，增长引入了重叠与陈旧，损害检索器在候选项间进行区分的能力 [179, 210]。因此，主动策管——包括合并冗余条目、剪除无效条目，以及对幸存技能进行分层组织——对于持续的性能至关重要。

技能治理（Skill Governance）。当技能被视作版本化的可执行包时 [178, 219]，它们便继承了传统软件的治理要求。新版本需要正确性测试，而随着环境变化，既有技能需要回归检查 [219]。跨平台分发进一步引入供应链风险：不正确或恶意的技能可能持续损坏智能体行为 [220, 177]。因此，大规模下的可靠运营要求版本控制、兼容性检查、冲突检测与有原则的弃用。

讨论

- 特异性与通用性之间存在一种根本张力：从轨迹中提取的技能往往狭窄而脆弱，而高度抽象的技能则缺乏可靠执行所需的操作精度。
- 一旦技能成为版本化产物，它们便继承软件生命周期负担——API 变更、环境迁移与静默错误传播——这要求主动维护，而不仅仅是提取。
- 记忆与技能之间的边界正在模糊：反复蒸馏的情景记忆变得与技能难以区分，这表明未来的框架应将经验累积、程序提取与技能维护统一进单一的受治理流水线。

3.4 多智能体编排

单智能体框架无论设计得多好，在并行性、专门化与容错方面都面临固有局限。多智能体编排通过协调多个智能体（每个智能体运行各自的工作流与记忆）朝向共享的任务目标来应对这些局限。产品实践识别出多智能体系统创造价值的三条轴线 [221]：上下文保护，即每个智能体在一个隔离的上下文窗口中运作；并行化，即以 token 消耗换取更短的墙钟时间；以及专门化，即按领域拆分智能体可避免单个智能体被加载过多工具时出现的性能退化。这一原则塑造了下文探讨的两个核心设计问题：定义智能体在结构上如何组织的协调架构，以及定义智能体在这些结构内如何交换信息的通信机制。

3.4.1 协调架构

协调架构定义了控制权、任务分配与结果聚合如何在各智能体间分布。现有方法依据系统是否依赖一个核心智能体来划分：中心化架构使用一个控制器来管理整个系统，而去中心化架构则通过对等智能体之间的交互实现协调。

中心化架构。中心化架构使用一个中心智能体来分解任务、将子任务分配给专用子智能体，并聚合它们的结果。MetaGPT [39] 与 ChatDev [40] 等早期系统通过手工设计的工作流组织角色专门化的智能体，但这些模式构建成本高且难以跨任务迁移。后续研究转向自主协调：AutoGen [38] 将调度视作一个可配置的对话与路由层，而 MegaAgent [222] 使智能体能在运行时协商职责，让组织无需预定义编排即可涌现 [223, 22, 224, 225]。生产中一个尤为成功的中心化模式是验证子智能体（verification subagent）[221]，即由一个专门的子智能体来测试或验证主智能体的工作。该模式之所以成功，是因为验证所需的上下文转移极少：验证器可以对系统进行黑盒测试而无需知道其完整构建历史，从而避免了推理在智能体间传递时发生的上下文退化。然而，扩展中心化系统并不像增加更多子智能体那样简单。近期发现 [226] 表明，配备精心设计记忆的小团队可以胜过更大的团队，这表明每个智能体的协调基础设施的质量比单纯的团队规模更重要。

去中心化架构。去中心化架构 [227] 探讨协作如何能在没有中心协调者的情况下从对等智能体之间的交互中涌现。AgentVerse [41] 证明智能体能够仅通过交互自发地形成角色与分工，DyLAN [228] 则以一个随任务推进而演化的动态协作图扩展了这一点 [42, 229, 230]。这种设计的一个已知风险是过早共识：如果智能体过快达成一致，少数但正确的观点可能被忽视。CONSENSAGENT [231] 通过分配非对称角色来解决此问题，使少数派立场在最终决策前获得结构化的考量。去中心化协调并不总是需要一个专门构建的通信协议。在一个大规模多智能体编码项目 [232] 中，16 个智能体在没有直接通信的情况下协调：每个智能体通过向共享目录写入一个文件来认领任务，冲突则通过 git 合并冲突

浮现出来。这表明既有的软件基础设施（如文件系统与版本控制）可以充当协调基底。共享状态模式 [233] 将此推广开来：智能体通过一个共享状态空间协作，在彼此的发现基础上构建而无需中心路由器，这消除了单点失败，但有重复工作与决策冲突的风险。

3.4.2 通信机制

通信机制处理智能体在多智能体系统内如何交换信息。没有精心设计的通信，即便结构良好的协调架构也会退化为彼此脱节的模型调用，而非一个连贯的协作系统。核心挑战在于决定共享什么信息以及以何种粒度共享。仅共享最终结果是不够的，因为动作携带了隐含的决策，缺乏结果背后推理的智能体很可能做出相互冲突的选择 [234]。共享完整的智能体轨迹保留了最大信息量，却造成令人却步的通信开销。一种实用的折中是将动作历史压缩为关键决策与事件的摘要，在控制成本的同时保持信息密度 [234]。现有研究从两个方向应对这一挑战：通过批判与分歧提升输出质量的基于辩论的方法，以及支持跨智能体协调执行的基于协作的方法。

基于辩论的方法。基于辩论的方法应对单个智能体的一个根本局限：沿着单一思路前进难以发现中间错误。一个自然的解法是让多个智能体从不同角色 [39, 235, 236] 与立场 [237, 238] 执行同一任务，使智能体在形成最终答案前相互挑战并精炼彼此的输出 [239]。然而，多智能体讨论并不会自动带来更好的结果 [240]。缺乏整合冲突观点的机制时，交互可能变得重复或不稳定。因此，更近期的工作围绕显式的反思循环 [241] 或面向共识的决策过程 [231] 来组织辩论，后者将冲突观点摘要并组装为一个连贯的结果。

基于协作的方法。基于协作的方法将智能体协调进一个连贯的执行结构，其中不同角色贡献互补的能力 [242, 243, 40]。早期研究依赖显式的角色专门化与预定义工作流 [244, 39]，使系统可控但在动态任务变化下脆弱 [223]。近期研究转向更自适应的方法，包括随执行演化而重构协调的基于图的规划 [245, 246]、联合优化协作策略的强化学习 [247, 248, 249, 250]，以及基于实时进度调整策略的环境感知协调 [251]。另一个新兴方向是以产物为中介的协作（artifact-mediated collaboration），即智能体通过持久的共享项目状态——如包含分析、规划与代码的权限范围化工作空间——而非短暂的对话式交接来协调 [252]。协作的一个实际挑战是生产系统中识别出的读写不对称 [253]：主要进行读取的多智能体系统远比主要进行写入的更易构建。当多个智能体同时写入时，它们隐含的决策可能冲突，产生难以合并的不兼容输出 [234]。这种不对称表明，协作式写入任务比并行式读取任务需要更强的协调协议。

讨论

- 每种协调模式都带有固有张力：中心化设计有单点瓶颈之险，而去中心化设计增加了调试与行为预测的难度 [233]。
- 分解应遵循上下文边界，而非问题类型：按角色拆分会造成持续的协调开销，而持有上下文的智能体应当拥有所有相关的子任务 [221]。
- 单个智能体的失败可能改变整条探索轨迹的方向，造成级联错误。
- 让一个多智能体系统基本运转起来约需 20% 的工作量，而使其可靠则需要剩下的约 80% [233]。
- 多智能体编排或许正在重走单智能体系统的老路：手工设计的协调模式最终可能被内化为模型的原生能力。

3.5 代表性智能体系统的对比视角

为使前述设计原则具体化，我们现在考察三个代表性智能体系统——Claude Code [88]、OpenClaw [11] 与 Hermes Agent [12]——如何在实践中实例化这四个框架组件。每个智能体系统都体现了一种独特的设计哲学，表 1 总结了关键差异。

表 1： 代表性智能体系统在四个框架组件上的设计原则。

组件	Claude Code	OpenClaw	Hermes Agent
智能体工作流	• 规划与执行分离 • 分层权限控制 • 交互式用户确认	• 声明式上下文装配 • 上下文驱动的工具选择 • 统一的多渠道路由	• 预算封顶的执行循环 • 安全感知的上下文过滤 • 平台自适应的提示设计
记忆系统	• 自动上下文压实 • 分层文件持久化 • 带类型的跨会话记忆	• 基于向量的语义检索 • 基于 wiki 的链接式知识 • 自动记忆捕获与召回	• 阈值触发的压缩 • 对话驱动的用户画像 • 基于快照的记忆加载
技能库	• 用户编写的斜杠命令 • 智能体辅助的技能创建 • 持久的定时任务	• 中心化技能注册表 • 标准化技能接口 • 基于角色的技能分配	• 智能体管理的生命周期控制 • 自动策管与退役 • 安全验证的激活
多智能体编排	• 基于角色的专家子智能体 • 仓库级工作空间隔离 • 前台与后台执行	• 深度受限的智能体衍生 • 基于身份的智能体配置 • 跨框架通信协议	• 显式的编排者角色 • 持久的跨智能体任务板 • 并行多模型执行

智能体工作流。三个系统以根本不同的方式掌管"感知、规划、动作"循环。Claude Code 强制执行严格的"先规划后执行"分离：智能体首先进入一个只读阶段以探索代码库并起草一份结构化规划，随后在执行任何修改前请求交互式用户确认，且每一次后续的工具调用都要经过分层权限控制机制，从而保证没有任何有后果的动作在缺乏显式同意的情况下进行。OpenClaw 将该循环视作可组合的基础设施，其中声明式上下文装配规定信息如何进入、留存于并离开上下文窗口，上下文驱动的工具选择基于认证状态或渠道类型等运行时信号在每次模型调用前剪枝可用工具集，而统一的多渠道路由让单个智能体能够通过一个分发接口服务于数十个消息平台。Hermes Agent 强调自主的自治，实现一个预算封顶的执行循环以显式的步数上限约束每个任务，安全感知的上下文过滤在每个被加载的文件到达模型前扫描其中的提示注入威胁，以及平台自适应的提示设计针对运行时环境裁剪系统指令，从而使智能体能够在无人监督下可靠运作。

记忆系统。三个系统在如何管理工作记忆与长期知识上差异显著。Claude Code 以上下文窗口为中心：对话本身充当工作记忆，一旦它变得过长，系统便执行自动上下文压实，将其压缩为结构化摘要；长期知识驻留在用户、项目与目录级别的分层文件持久化中，并被自动加载进每个会话，辅以一个带类型的跨会话记忆存储，按用户偏好、反馈、项目状态等类别组织持久事实。OpenClaw 采取双轨设计，将在每次模型调用前浮现相关事实的基于向量的语义检索，与维护人类可读、相互链接的持久文章的基于 wiki 的链接式知识配对，二者都由自动记忆捕获与召回支撑，后者记录值得注意的事实并在无需显式用户命令的情况下检索它们。Hermes Agent 专注于流式上下文管理：阈值触发的压缩实时监控 token 用量，一旦越过可配置的上限便压实上下文，同时保护最早与最近的消息，使任务设定与近期状态得以幸存；对于长期用户建模，它通过一个外部服务采用对话驱动的用户画像，该服务

通过对过往对话进行推理来维护一个不断演化的用户画像，并采用基于快照的记忆加载，在会话开始时将记忆文件作为冻结快照摄入，从而使会话中途的更新不会使提示缓存失效。

技能库。三个系统以递增的自主程度组织可复用能力。Claude Code 将技能视作用户编写的斜杠命令，每个技能是一个带结构化元数据的 markdown 文件；一个内置助手支持智能体辅助的技能创建，使系统能够自行引导出新技能，而持久的定时任务让周期性作业表现得像持久技能，所有这些都受插件黑名单与细粒度权限白名单的治理。OpenClaw 转向一个中心化技能注册表，将每个技能与标准化技能接口打包，并通过一个中心枢纽发布以供发现与安装，基于角色的技能分配确保每个智能体只接收与其指定角色相关的技能，将技能的编写与消费干净地分离。Hermes Agent 走得最远，实现了智能体管理的生命周期控制：技能通过三个细节递增层级的渐进式披露被暴露，使智能体只加载所需的内容，自动策管与退役移除在可配置周期内未被调用的技能，而安全验证的激活在每个技能被启用前对其加以校验。

多智能体编排。三个系统都支持子智能体委派，但在协调哲学上有所不同。Claude Code 遵循基于角色的专家子智能体模式，衍生出带有显式角色注解（如探索者、规划者或通用型）的智能体，每个都接收一个受约束的工具集；仓库级工作空间隔离为每个子智能体提供其自身的工作副本，而前台与后台执行的选项使每个子智能体保持在其自身干净的上下文窗口中。OpenClaw 采取协议优先的立场，以深度受限的智能体衍生约束递归以防止失控链条，以基于身份的智能体配置通过范围化配置文件定义每个智能体的角色、个性与行为约束，并将一个跨框架通信协议作为一等关切，使构建在不同技术栈上的智能体能够互操作。Hermes Agent 提供了最丰富的协调工具包，定义一个显式的编排者角色以将协调型智能体与执行任务的叶子智能体分开，维护一个持久的跨智能体任务板以共享工作跟踪，并且最为独特地支持并行多模型执行，即同时查询多个语言模型并综合它们的输出。

> **讨论**
>
> - 同样的四个框架组件可以在差异极大的设计哲学下被实例化：Claude Code 为人在回路（human-in-the-loop）的安全而优化，OpenClaw 为可组合的可扩展性而优化，Hermes Agent 则为自主的自治而优化。
> - 这些权衡直接反映部署语境：交互式开发者辅助、多渠道平台编排与自主终端执行，要求在控制、灵活性与自主性之间取得根本不同的平衡。

4 面向框架的模型适配

在框架工程的协同框架内，设计外围框架必须辅以模型侧的适配。标准 LLM 是为通用文本生成而训练的，缺乏调用工具、遵循多步工作流或处理环境反馈的内置能力。我们将模型适配组织为两类：上下文工程（第 4.1 节），它通过框架所暴露的信息在推理时塑造模型行为；以及智能体训练（第 4.2 节），它通过模仿与强化学习将智能体特定的行为内化进模型参数。

4.1 上下文工程

推理期间提供的上下文信息决定了一个 LLM 智能体的行为及其能力上限。如第 3 节所讨论，框架掌管模型在每一步观察到的内容，包括任务指令、工具描述、检索到的知识与交互历史。上下文工程通过决定暴露什么信息、如何组织它，以及如何在多轮交互中更新它，来适配模型行为。与一般提示不同，面向智能体框架的上下文工程专门针对有限上下文预算下的工具使用、工作流执行与长程决策。我们涵盖两个互补的阶段：在模型行动前构建高质量输入的上下文设计，以及在长时交互中压缩并维护上下文状态的上下文管理。

4.1.1 上下文设计

上下文设计聚焦于在智能体推理或行动前构建最优的输入。这依赖两个核心策略：组织提供给模型的指令的提示工程，以及通过纳入外部信息扩展模型知识的上下文检索。

提示工程（Prompt Engineering）。提示工程通过精心设计的输入、在不更新参数的情况下引导模型行为。早期研究聚焦于组织任务指令与示范 [254, 255, 125, 126]，而后续研究表明显式的角色描述能够有效引出有针对性的行为 [256, 227]。这在智能体承担专门角色并在共享工作流中协调的多智能体系统中尤为有用 [159, 45]。手工提示设计的一个常见局限是它严重依赖人力。为解决这一问题，自动化提示优化使模型能够精炼自身的提示。APE [16] 使用 LLM 自身来生成并选择候选指令，而后续方法通过自我精炼循环彻底消除了人工干预 [257, 258, 259]。

上下文检索（Context Retrieval）。上下文检索动态整合外部知识，以克服参数化记忆的局限。传统的检索增强生成（RAG）依赖于以平铺文本块进行一次性检索的静态范式，难以应对复杂的多跳查询 [260]。后续方法沿两个维度推进检索：检索时机与知识结构。在检索时机方面，主动机制允许模型将推理与信息收集交织起来，基于中间状态动态决定何时检索以及如何重写查询 [261, 262, 263, 264, 265, 266, 44]。在知识结构方面，基于图或表的索引捕捉检索项之间的显式关系 [267, 268, 269, 43]。在这些进展之上，智能体式检索将检索操作直接整合进模型的迭代执行循环。由 ReAct [82] 开创的这一范式使智能体能够在推理与查询之间交替。后续框架进一步将检索抽象为自动构建的模块 [270] 与分层策略 [271]，而多智能体生态系统则将检索分布到专门角色上，通过协作推理来处理复杂任务 [272]。

4.1.2 上下文管理

原始上下文往往充满噪声且过于冗长，既有上下文溢出之险，又有推理退化之虞 [273]。上下文管理涉及两个过程：为即时推理压缩信息的上下文处理，以及跨交互维护累积经验的上下文更新。

上下文处理（Context Processing）。上下文处理压缩已装配的上下文，以在模型有限的窗口内维持最大的信息密度。早期方法基于重要性剪除 token 来缩短输入文本 [274, 275, 276]。这些方法对单轮查询有效，却忽视了自主智能体的序列动态。为防止长程任务中的上下文溢出，近期框架专门为多轮智能体轨迹设计压缩。Acon [277] 在交互历史上应用强化学习，以在维持长期记忆的同时最小化上下文用量。AgentDiet [46] 剪除冗余信息以降低推理成本，AgentFold [278] 通过分层合并凝练子任务，而 SWE-Pruner [279] 为编码智能体执行任务感知的自适应剪枝。在剪除既有轨迹之外，IterResearch [280, 281] 将上下文管理重新表述为工作空间重建，用一个由原始问题、不断演化的报告与最新动作–观测对组成的紧凑状态取代完整历史的重放。

上下文更新（Context Updating）。上下文更新在长时交互中动态维护并演化一个智能体的上下文状态，决定记住什么、遗忘什么，以及如何重组经验。在基础层面，智能体通过反思累积经验。Reflexion [24] 开创了这一点，让智能体为未来的尝试总结失败原因，而 AWM [146] 从成功轨迹中蒸馏可复用的工作流模式。随着经验增长，平铺存储变得不切实际，从而催生了分层记忆架构 [148]，将记忆划分为一个快速的上下文窗口与一个具有自主页面交换的慢速外部存储。后续框架以图数据库 [29]、流式缓存 [282] 与时间衰减遗忘 [30] 增强这一点。然而，仅靠被动存储无法支持主动的记忆生命周期管理。近期研究赋予智能体组织相互关联知识 [32, 283, 284]、突显重要记忆 [31] 与删除过时信息 [285, 286] 的能力。

4.2 智能体训练

上下文工程在推理时适配模型行为，但由于参数保持固定，其有效性受限于模型的内在能力。智能体训练对此加以补充，通过两大范式将领域知识与智能体特定行为直接内化进模型参数：从示范或教师生成的轨迹中学习的监督微调（SFT），以及通过环境交互反馈精炼策略的强化学习（RL）。将这些方法应用于智能体引入了独特挑战：构建交互式环境、在稀疏奖励下分配信用（credit）、稳定策略优化，以及管理大规模工程开销。我们讨论四个方面：环境构建（第 4.2.1 节）探讨如何构建交互式训练环境；奖励设计（第 4.2.3 节）考察如何获取稳定的奖励信号；训练优化算法（第 4.2.2 节）讨论基于 SFT 与基于 RL 的策略更新；基础设施（第 4.2.4 节）讨论大规模分布式训练的系统。

4.2.1 环境构建

训练环境构成智能体训练的基础。由于框架居中处理模型与外部世界之间的交互，环境保真度决定了所学策略能否迁移到现实世界的部署。现有环境依据反馈机制分为三种类型：基于规则的环境从执行结果提供确定性的二元奖励；基于仿真的环境使用代理模型来近似现实世界的动态；现实世界环境则将智能体直接置于具有真实反馈的活系统中。

基于规则的环境。基于规则的环境已从预定义沙箱演进到严格可验证的指标。早期研究 [287] 在具有预定义状态转移的基于规则的沙箱中训练智能体，支持家庭 [288] 或电子商务 [289] 场景。然而，预定义动作需要手工设计，且难以泛化到具有大动作空间的复杂推理。近期研究 [290, 291, 292, 293] 表明，借助可靠的二元结果反馈，智能体能够在无人工标注的情况下持续自我改进。这一原则在数学 [294, 295, 10] 与编码 [296, 132] 等由精确验证器提供可靠反馈的领域被迅速采用。更近期的环境通过程序化生成的推理任务 [297] 与支持长程规划的有状态企业沙箱 [298] 进一步扩展这一点。在

软件工程中，Scale-SWE [299] 通过将原始 GitHub 拉取请求转化为经验证的、Docker 化的 SWE 任务，并经由沙箱化的多智能体工作流生成训练轨迹，来扩展环境构建。

基于仿真的环境。基于仿真的环境使用代理模型提供近似现实世界动态的低成本反馈。一个关键挑战是"LLM 作为仿真器"的方法在物理场景中易受幻觉影响。为提升保真度，一些方法 [300] 在大规模交互数据上训练世界模型以实现跨域泛化，而神经仿真器通过建模屏幕转移将其扩展到操作系统接口 [301]。在保真度之外，提升 rollout 效率同样重要。硬件加速的物理引擎支持大规模并行的多智能体 rollout [302, 303]，而 RAP [304] 将仿真转移到模型本身内部，使其在行动前比较可能的结果。为进一步提升准确性，近期研究 [305] 用以数据库为后端、将动作作为 SQL 事务执行的确定性机制取代生成式代理。

现实世界环境。现实世界环境将智能体直接置于具有真实、不断变化反馈的活数字或物理系统中。早期研究 [108, 306, 307] 使用受控的静态网站，但这些缺乏真实感。后来的研究引入了实际的数字噪声，如网络延迟与动态内容加载 [113, 308, 309, 310]。为进一步扩大规模，一些研究 [311, 312, 313] 让智能体使用实时操作日志作为反馈自由浏览活网络。物理世界是最复杂的环境，因为机器人动作必须遵循真实物理并服从严格的安全规则。具身环境 [314, 315, 316] 将实时传感器数据与基于物理的安全检查相结合，使智能体能够将高层推理翻译为真实的物理运动。

4.2.2 训练优化算法

在构建了交互环境与奖励信号之后，智能体训练需要优化算法来更新模型参数。现有方法遵循两大范式：通过模仿示范或自生成轨迹来学习智能体行为的监督微调，以及通过与环境反馈的试错交互来精炼策略的强化学习。

监督微调（Supervised Fine-Tuning）。监督微调通过在包含中间规划、工具调用与环境观测的多轮轨迹上直接优化，将智能体能力迁移进目标模型。依据数据来源，方法分为来自教师模型的离策略（off-policy）蒸馏与来自学生自身的在策略（on-policy）生成。离策略蒸馏在更强模型预先生成的轨迹上训练。FireAct [317] 首次将其应用于智能体场景，表明微调后的小模型在工具使用任务上能胜过更大的模型。由于仅含智能体的数据可能削弱通用能力，后来的研究通过轨迹—指令混合 [318]、格式与推理信号解耦 [319] 以及统一的多轮格式化 [320] 来提升数据质量。为提升可扩展性，近期研究通过环境交互与系统化数据工程合成训练数据 [119, 321, 322, 323, 299]。在这条研究脉络中，ClawGym [324] 为 Claw 风格的个人智能体引入了一个以合成为中心的可扩展框架，生成以工作空间为依托的任务，并进一步将其与智能体训练和诊断性基准构建相连。当轨迹同时包含规划与执行信号时，多教师监督 [325, 326] 或片段特定损失 [327] 可分离这些成分以实现更好的迁移。在策略蒸馏通过让学生生成其自身轨迹、同时由教师提供稠密监督，解决了训练—测试分布失配问题。早期方法优化诸如正向 KL [328] 与反向 KL [329] 之类的散度目标。为在能力差距较大时稳定训练，后续方法调整目标 logits [330] 或裁剪奖励信号 [331]。在模仿之外，ExOPD [332] 将蒸馏与 KL 正则化的 RL 结合，以将学生推到教师之上，而 GAD [333] 使用基于判别器的反馈将在策略蒸馏扩展到黑盒教师。

强化学习方法。监督微调内化了智能体行为，却无法支持超出训练分布的探索。强化学习通过带奖励反馈的试错来精炼策略以解决这一问题。依据是否使用价值网络，方法分为基于评论家（critic-based）与无评论家（critic-free）的算法。基于评论家的算法引入一个价值网络来估计优势

（advantage）并引导稳定的策略更新。PPO [334] 是代表性方法，其改进包括价值预训练 [335] 与奖励归一化 [336, 337]。然而，标准 PPO 在长程智能体上表现欠佳，因为跨多步轨迹的信用分配本质上很困难。近期方法通过将交互分解为规划层级 [338, 339] 或推理树 [340]，以及在多智能体设定下为不同智能体分配独立评论家 [341] 来解决此问题。无评论家的算法移除价值网络以降低内存与计算成本。一条脉络使用离线偏好学习，如 DPO [342]、KTO [343]、SimPO [344] 与 ORPO [345]，以避免昂贵的在线探索。另一条则保留在线探索，但直接从奖励估计优势：GRPO [346] 与 ReMax [347] 使用基于组级或最大回报信号的无参数估计器。对于智能体工作负载，近期研究进一步引入分层采样与树状探索 [50, 348, 349]、token 级损失加权 [48]、动态采样 [350, 351] 与熵平衡 [351, 352, 353]，以稳定长程探索。

4.2.3 监督信号

在环境构建之后，奖励信号评估智能体行为并指导策略改进。依据监督粒度，出现了两种互补的范式：结果级信号仅在任务完成时提供反馈，而过程级信号则在每一步后提供稠密监督。

结果级信号（Outcome-Level Signals）。结果级信号评估任务的最终状态，在那些易于验证却难以求解的领域中表现良好。在数学 [354, 355]、代码生成 [356, 357]、软件工程 [358] 与深度研究 [49, 312, 359] 任务中，客观的执行反馈充当一种天然的结果信号。一种常见做法 [358, 360] 使用二元信号让模型从正确与错误的轨迹中学习，驱动大规模自博弈与迭代修复。为进一步减少对人工标签的依赖，TTRL [361] 通过对采样解进行多数投票来估计奖励，使无金标注的 RL 成为可能。在信号极其稀疏的形式化定理证明中，DeepSeek-Prover-v1.5 [362] 将证明助手反馈与基于树的搜索及探索奖励相结合。

过程级奖励（Process-Level Rewards）。结果级奖励在长程任务中受困于信号稀疏，因为模型无法确定具体哪些步骤导致了成功或失败。过程奖励模型（PRM）通过在每个推理或动作步骤提供稠密监督来解决这一问题。PRM 的演进涉及两个方面：监督信号构建与架构设计。在信号构建方面，早期方法 [363] 依赖昂贵的人工标注。后续研究通过蒙特卡洛采样 [364] 与蒙特卡洛树搜索 [365] 将其自动化。近期扩展为 GUI 定位引入连续的空间奖励 [366]，并为工具选择引入细粒度奖励 [51]。在架构设计方面，传统 PRM 是为每一步分配单一分数的判别式模型 [363]，但它们难以解释且易产生偏差。生成式方法 [367] 通过自然语言推理执行步骤验证、用思维链解释正确性来解决此问题，产生更透明的奖励信号。作为基于奖励的优化的补充，在经策管的轨迹上进行监督微调提供了一种更直接的信号：高质量轨迹源自人类专家 [368]，或经由拒绝采样从强模型中蒸馏 [258]，许多系统将二者结合以引导出能力 [4]。

4.2.4 基础设施

大规模智能体训练要求对模型推理、环境交互、梯度计算与参数同步进行协调调度。系统基础设施的效率直接限制了优化的速度与可扩展性。

通用框架。通用智能体框架在分布式集群上协调核心 RL 组件（actor、critic、奖励模型与参考模型），同时支持多轮交互与持续的环境反馈。为实现灵活调度，一些系统 [369, 370, 47, 371] 动态地将模型分配到特定的 GPU 分区，优化跨训练阶段的数据传输。为提升吞吐量，异步架构 [372, 373] 将较慢的生成阶段与梯度更新分开，可高效扩展到数千块 GPU。为实现环境集成，标准化 API [374,

375] 将各种仿真器包裹进统一接口以进行跨环境训练。然而，智能体的多轮特性导致在快速生成、缓慢环境交互与繁重梯度更新之间的负载不均衡。近期框架通过在单个工作流内交织生成与环境执行 [50, 348, 376, 377, 378, 379]，或将各阶段在机器间物理分离 [380, 381] 来解决这一问题。

专用框架。专用框架为多智能体协调与全模态交互等高级设定扩展通用基础设施。对于多智能体系统，多个智能体通过长程、相互依赖的轨迹交互，使 rollout 调度远比单智能体设定更难 [382]。专用框架 [383, 382, 341, 384, 385] 引入中心化的多智能体交互、分布式策略训练与异步多轮 rollout。对于全模态智能体训练，跨文本、图像与视频的异构观测与反馈带来独特挑战 [386]。近期研究为多模态锚定 [386, 387] 与多模态搜索环境 [388] 引入专门的评分标准（rubric）奖励，将智能体训练扩展到标准的纯文本单智能体设定之外。

5 框架能力与评测

在这一部分，我们回顾针对智能体框架的代表性基准与评测方法。一个智能体系统的能力不仅取决于底层模型，也同样取决于组织上下文、工具、执行与反馈的框架。我们将这种在工具使用、部分可观测性、长执行轨迹与变化的环境反馈下维持有效行为的整体能力，称为**框架能力（harness capacity）**。度量框架能力需要超越单步正确性；评测必须评估一个系统能否在与外部环境的多步交互中维持目标导向的进展。近期基准通过将智能体置于日益真实的环境中、延长任务时域与有状态性，并纳入更丰富的评测信号（包括执行结果、环境状态、基于评分标准的判断与多次运行的可靠性），来体现这一视角 [108, 389, 105, 390, 132, 53, 62, 60, 73, 391, 392, 59, 393, 68, 394]。因此，基准设计本身已成为框架工程不可或缺的一部分，因为它决定了编排的哪些方面——上下文管理、工具协调、错误恢复——能够被可见化与可度量化。

表 2：智能体框架工程中跨主要任务领域的代表性基准。("评测机制"中：基于规则 = Rule-based；LLM 评判 = LLM Judge；人工评判 = Human Judge)

基准	日期	任务	评测机制
深度研究（Deep Research）			
BrowseComp	2025.04	基于浏览的信息搜寻	基于规则
Deep Research Bench	2025.05	网络研究	基于规则
DeepResearch Bench	2025.06	深度研究报告生成	基于规则
ResearcherBench	2025.07	前沿科学探究	基于规则 / LLM 评判
ReportBench	2025.08	学术综述 / 报告生成	基于规则
Characterizing Deep Research	2025.08	深度研究基准刻画	基于规则
WideSearch	2025.08	智能体式广域信息搜寻	基于规则
PDR-Bench	2025.09	个性化深度研究	LLM 评判
LiveResearchBench	2025.10	真实场景中以用户为中心的深度研究	LLM 评判
ResearchRubrics	2025.11	基于提示与评分标准的深度研究评测	LLM 评判 / 人工评判
DEER	2025.12	深度研究专家报告	LLM 评判
DeepSynth-Eval	2026.01	深度综述写作中的信息整合	基于规则
VideoDR-Benchmark	2026.01	开放网络上的视频深度研究	基于规则
DeepResearch Bench II	2026.01	基于评分标准的深度研究报告诊断	基于规则
MMDeepResearch-Bench	2026.01	多模态深度研究	基于规则 / LLM 评判
DeepSurvey-Bench	2026.01	所生成科学综述的学术价值	人工评判
DeepSearchQA	2026.01	穷尽式答案集生成	基于规则 / LLM 评判
deepsynth-bench	2026.02	深度信息综合	基于规则 / LLM 评判
软件工程（Software Engineering）			
RepoBench	2023.06	仓库级代码自动补全	基于规则
SWE-bench	2023.10	真实世界 GitHub issue 解决	基于规则
CodeAgent	2024.01	工具集成的仓库级编码挑战	基于规则
LiveCodeBench	2024.03	整体的、无污染的代码评测	基于规则
DevEval	2024.05	与仓库对齐的代码生成	基于规则
BigCodeBench	2024.06	函数调用与复杂代码生成	基于规则
REPOCOD	2024.10	真实的仓库级编码	基于规则
HumanEval Pro / MBPP Pro	2024.12	自调用代码生成	基于规则
Commit0	2024.12	从零开始的库生成	基于规则
FEA-Bench	2025.03	仓库级特性实现	基于规则

基准	日期	任务	评测机制
AutoCodeBench	2025.08	自动化代码基准生成	基于规则
SWE-Bench Pro	2025.09	长程软件工程任务	基于规则
SWE-Sharp-Bench	2025.11	可复现的 C# 软件工程任务	基于规则
LoCoBench-Agent	2025.11	长上下文软件工程	基于规则
NL2Repo-Bench	2025.12	长程仓库生成	基于规则
OmniCode	2026.02	多样化的软件工程智能体任务	基于规则
SWE-Universe	2026.02	可扩展的可验证软件环境	基于规则
FeatureBench	2026.02	复杂特性开发	基于规则
BeyondSWE	2026.03	超越单仓库缺陷修复	基于规则
工具使用与函数调用（Tool Use and Function Calling）			
API-Bank	2023.04	API 规划与工具调用	基于规则
ToolLLM / ToolBench	2023.07	大规模真实世界工具使用	LLM 评判
AgentBench	2023.08	多领域交互式智能体	基于规则
GAIA	2023.11	通用 AI 助手任务	基于规则
τ-bench	2024.06	工具—智能体—用户交互	基于规则
AppWorld	2024.07	可控的应用与人物世界	基于规则
AssistantBench	2024.07	真实的网络助手任务	基于规则
ToolSandbox	2024.08	有状态的对话式工具使用	基于规则
ToolHop	2025.01	查询驱动的多跳工具使用	基于规则
DICE-BENCH	2025.06	多轮、多方对话工具使用	基于规则
τ^2-Bench	2025.06	双控设定下的对话式智能体	基于规则
BFCL	2025.07	函数调用与智能体评测	基于规则
FuncBenchGen	2025.09	可控的多步函数调用	基于规则
GAIA2	2025.09	真实生活助手任务	基于规则
CCTU	2026.03	复杂约束下的工具使用	基于规则
计算机使用与 GUI 定位（Computer Use and GUI Grounding）			
Mind2Web	2023.06	通用型网络智能体交互	基于规则
WebArena	2023.07	真实的网络导航	基于规则
Android in the Wild	2023.07	Android 设备控制	基于规则
VisualWebArena	2024.01	多模态视觉网络任务	基于规则
WorkArena	2024.03	常见知识工作类网络任务	基于规则
OSWorld	2024.04	开放式真实计算机使用	基于规则

基准	日期	任务	评测机制
AndroidWorld	2024.05	动态 Android 交互环境	基于规则
Windows Agent Arena	2024.09	多模态桌面操作系统任务	基于规则
WorldGUI	2025.02	桌面 GUI 自动化的动态测试	基于规则
ScreenSpot-Pro	2025.04	专业级高分辨率 GUI 定位	基于规则
MMBench-GUI	2025.07	分层的多平台 GUI 评测	基于规则
OSWorld-MCP	2025.10	计算机使用智能体中的 MCP 工具调用	基于规则
VenusBench-GD	2025.12	多平台 GUI 定位	基于规则
机器学习工程与科学研究（ML Engineering and Scientific Research）			
DS-1000	2022.11	数据科学代码生成	基于规则
MLAgentBench	2023.10	机器学习实验	基于规则
DA-Code	2024.10	智能体数据科学代码生成	基于规则
ScienceAgentBench	2024.10	数据驱动的科学发现	基于规则
MLE-bench	2024.10	机器学习工程	基于规则
PaperBench	2025.04	AI 研究复现	LLM 评判
TimeSeriesGym	2025.05	时间序列机器学习工程	基于规则
MLR-Bench	2025.05	开放式机器学习研究	LLM 评判
FML-bench	2025.10	自动化机器学习研究	基于规则
TRAJECT-Bench	2025.10	智能体工具使用的轨迹感知评测	基于规则 / LLM 评判
ReplicatorBench	2026.02	社会与行为科学中的可复现性	LLM 评判 / 人工评判

5.1 深度研究

深度研究基准评估一个智能体能否支持超越简答检索的开放式信息搜寻。该领域已从困难的浏览类问题，进展到更广域的搜索、有引证支撑的报告撰写，以及基于评分标准的综合质量评测。

广域与复杂搜索。一个早期方向聚焦于信息密集型的搜索，而非简答查找。BrowseComp 与 WideSearch 评估一个智能体能否维持多步探索、跨搜索分支比较证据，并保留有用的中间发现 [52, 53, 54, 391]。LiveDRBench 尤为值得注意，它以广度、逻辑嵌套与搜索强度（而非仅以报告长度）来刻画深度研究。与经典检索设定相比，这些基准着重考验搜索规划、证据累积，以及框架在搜索空间扩展时维持覆盖率的能力。

研究报告。第二条研究脉络将关注点从信息收集转向有引证支撑的长篇报告撰写。DeepResearch Bench、ReportBench 与 LiveResearchBench 评估一个智能体能否将检索到的证据组织为连贯的输出，其论断、引证与结构均可被直接检视 [55, 395, 396, 392, 397, 398, 399]。这一方向更看重报告组织、引证质量，以及陈述与来源之间的忠实度，揭示出性能在多大程度上依赖于框架在生成期间管理笔记、压缩文档与保持内容锚定的能力。

综述生成与评测。更近期的基准转向带显式质量评测的综述式综合。ResearchRubrics 将深度研究提示与专家设计的评分标准配对，使覆盖度、锚定性与分析质量成为一等的评测目标 [400, 392, 397]。这标志着一种转变：从询问一个智能体能否产出一个貌似合理的产物，转向询问其综合在结构化判断下是否完整、是否有充分支撑、是否真正有用。即便如此，当前的深度研究基准仍对来源可信度、网络内容变化下的引证忠实度、跨会话持久化，以及实用研究助手中至关重要的成本–延迟权衡评估不足。

5.2 软件工程

软件工程基准评估一个智能体能否在真实的开发条件下运作，而不仅仅是生成独立的代码。评测已从仓库理解与 issue 修复，扩展到特性实现与更广泛的工程工作流。

仓库任务。一个早期方向聚焦于以仓库为依托的编码，而非孤立的生成。这些任务考察一个智能体能否处理项目结构、跨文件依赖与文档，而不仅仅依赖一个局部提示。RepoBench 代表面向检索与补全的评测，而 CodeAgentBench 与 REPOCOD 则转向带工具使用与可执行评测的真实仓库级生成 [401, 402, 56, 403]。仓库任务主要评估代码库锚定、上下文选择与导航，为后续的 issue 解决与工程基准奠定基础。

Issue 解决。第二条脉络聚焦于可执行的修复与 issue 级调试。SWE-bench 将真实 issue 解决作为核心任务，SWE-Bench Pro 将其扩展到更长程、多文件、企业级的问题，而 LiveCodeBench 则增加了可执行的、污染感知的评测 [404, 57, 60, 58, 405]。这一方向强调迭代纠错、调试与响应测试反馈。随着任务变得更长程，性能越来越取决于框架能否检索上下文、运行命令、检视失败并支持反复修复。

更广泛的工程工作流。第三条脉络将评测从 issue 修复拓展到更完整的工程工作流。NL2Repo-Bench 要求智能体设计架构、管理依赖，并从自然语言需求生成一个可安装的仓库 [406]。FeatureBench 聚焦于需要跨文件协调编辑的特性级实现 [407]。BeyondSWE 整合了多种"超越缺陷修复"的范式，包括跨仓库推理、领域专门化修复、依赖迁移与文档到仓库的生成 [408, 59]。这些基准共同表明，更新的评测越来越多地考验全局规划、跨文件一致性与依赖管理，而非仅仅局部补丁的生成。

5.3 工具使用与函数调用

工具使用与函数调用基准评估一个智能体能否正确地、相关地、并与任务状态一致地调用外部能力。整体趋势从孤立的函数调用，转向多步、有状态、对话式的工具编排。

函数调用。一个早期方向孤立地考察调用本身的正确性。API-Bank 与 BFCL 评估一个系统能否选择合适的函数、提供有效的参数，并使调用与任务要求对齐 [409, 62, 393]。这针对的是框架能力中一个狭窄但基础的层面：在更长轨迹开始之前选择并格式化正确的动作。

工具使用。第二条脉络将范围从调用一个工具拓展到用工具解决一个任务。ToolBench、AgentBench 与 GAIA 评估智能体能否在 API 上进行规划、整合中间结果，并将工具作为更广泛任务完成的一部分来使用 [115, 307, 410, 61]。这些任务强调工具路由、多步规划，以及将检索到的输出与后续决策相连的能力，使评测从局部调用正确性转向协调式的、工具支撑的问题求解。

有状态交互。更近期的基准转向跨多轮与变化状态的持久编排。τ-bench 与 AssistantBench 评估用户约束下的对话式交互，而 AppWorld、ToolSandbox 与 τ^2-Bench 则要求在更长的轨迹中管理共享的应用或工具状态 [390, 132, 411, 64, 65, 63, 412]。这些基准更看重状态跟踪、多轮一致性，以及依赖动作的正确排序，尽管它们在权限、安全边界、部分失败恢复与跨会话记忆方面仍留有缺口。

5.4 计算机使用与 GUI 定位

计算机使用与 GUI 定位基准评估一个智能体能否在浏览器、桌面与移动系统等交互式数字环境中感知并行动。近期基准越来越强调多模态感知、更丰富的环境，以及显式的 GUI 定位。

网络导航。一个早期方向是基于浏览器的任务完成。Mind2Web 与 WebArena 通过要求系统解读页面内容、选择动作，并在真实的网络状态序列中取得进展，将网络导航确立为一个核心的评测设定 [107, 108]。这些任务强调界面内部的观测、动作选择与轨迹管理，将浏览器引入为评估智能体交互的一个有意义的环境。

计算机使用。第二条脉络从网站扩展到更广泛的数字环境。VisualWebArena 增加了视觉锚定的网络交互，而 WorkArena、Android in the Wild、OSWorld、AndroidWorld 与 Windows Agent Arena 则将评测扩展到企业平台、移动任务、桌面系统与真实的 Windows 控制 [104, 389, 413, 105, 69, 414, 67]。这些基准更强调多模态感知、环境状态与更广泛的计算机控制，使评测更贴近实用智能体被部署的条件。

GUI 定位。更近期的工作转向将动作显式定位到图形界面中。ScreenSpot-Pro 与 OSWorld-MCP 将元素识别与界面对齐的控制作为任务的直接组成部分，而非隐含的子问题 [415, 70, 416, 68]。这反映出向通用多模态计算机使用的转变，其中精确定位成为可靠动作的核心瓶颈。即便如此，沙箱设定与实时部署之间仍存在可观的差距，而干预点、安全敏感动作、延迟与实时部署差距等问题仍只得到微弱的评估。

5.5 机器学习工程与科学研究

这一类别的基准评估一个智能体能否在比普通编码任务更长的时域上支持实验、工程与科学工作流。其发展从数据科学问题求解，转向研究复现与开放式研究辅助。

机器学习实验。一个早期方向聚焦于数据科学与机器学习中的可执行实验。DS-1000 评估真实的数据科学编码，而 MLAgentBench 则考察智能体能否迭代地运行并改进实验，而不仅仅生成代码 [417, 418, 394]。这一设定强调实验设置、数据处理与经验反馈，度量框架能否支持一个实验循环，而非单一的生成步骤。

科学工作流。在可执行实验的基础上，第二条脉络将评测拓展到面向工作流的科学与机器学习工程任务。DA-Code、ScienceAgentBench 与 MLE-bench 在相互依赖的步骤上强调更广泛的数据科学工作流、科学任务分解与机器学习工程流水线 [419, 10, 72, 71]。性能越来越取决于框架能否在长时工作流上组织上下文、执行与验证。

研究复现。更近期的基准转向复现与开放式科学验证。PaperBench、MLR-Bench、FML-bench 与 ReplicatorBench 评估智能体能否在更长程的条件下复现研究结果、开展开放式机器学习研究，并

重建科学论断 [73, 420, 421, 422]。这代表着一种更广泛的转变，即转向那些询问系统能否在长时轨迹上支持有意义科学工作的基准。科学上的有用性仍比工程正确性更难验证，有效性、新颖性、算力不平等与长程可靠性都仍是开放挑战。

讨论

- 基准设计正从单信号的最终答案检查，演变为多信号评测：基于执行的、基于状态的、基于评分标准的与轨迹感知的信号共同构成一条从"验证产出了什么"到"评估它是如何产出的"的递进。
- 关键缺口依然存在：安全与权限控制、运营成本（延迟、token 预算）与长期记忆持久化，在当前各基准中都被微弱或不一致地评估。
- 最根本地，当前基准很少将归因于基座模型的改进与归因于框架的改进区分开来，限制了框架工程师可付诸行动的洞见。
- 弥合这些缺口——尤其是成本感知评测、安全测试与模型–框架归因——将是下一代基准的核心。

6 未来方向

智能体框架的发展正从构建有能力的执行循环，转向治理长时运行的运行时系统。早期框架设计聚焦于使模型能够感知环境、调用工具、维护上下文与协调动作。随着这些能力成熟，核心瓶颈转移到了：当智能体在延长的时域上运行、与不受控环境交互、跨会话持久化状态，并以日益增大的权限行动时，会发生什么。在此类设定中，性能不仅取决于模型能否产生一个貌似合理的下一动作，也取决于框架能否选择性地分配算力、管理状态生命周期、维护与环境对齐的信念、约束高风险动作，并为评测暴露证据。这一转变将未来的框架工程刻画为在资源、状态与动作约束下的运行时治理。效率关注算力、上下文与工具如何在显式预算下分配。安全关注在模型输出成为现实世界动作之前，工具访问、记忆写入与委派如何被居间管控。持续学习关注持久记忆与技能如何被巩固、更新并免于漂移。状态与环境建模关注部分观测与工具反馈如何被维护为连贯的信念状态。具身框架在动作充满噪声且常不可逆的物理约束下检验这些抽象。评测则必须超越最终成功率，转向轨迹感知、安全感知与成本感知的协议。以下各小节依次讨论这些方向。

6.1 效率

随着 LLM 智能体从单轮生成转向长程执行，效率成为一个框架级而非仅模型级的关切。智能体系统中的成本通过反复的模型调用、不断扩张的动作—观测历史、工具结果反馈、检索、反思与多智能体委派而累积。一个对每个子任务都施加相同模型、相同上下文预算与相同推理深度的统一执行循环，本质上是浪费的，然而当前大多数框架恰恰采用这种统一策略。模型路由工作提供了第一种替代方案，它将推理表述为成本—质量权衡，表明可以动态地选择较弱与较强的模型，而不必总是调用能力最强的模型 [423]。同样的原则也延伸到推理深度：对思维链提示的分析告诫说，更深的推理轨迹并非普遍有益，因为当额外的推理无法泛化时，它可能脆弱、任务特定且代价高昂 [424]。综合来看，这些结果表明智能体效率应被作为一个调度问题来研究：框架必须决定何时值得进行更深的推理、使用更强的模型或检索，以及何时更便宜的执行路径就已足够。

模型选择与推理深度决定了每个单独步骤的成本，而第二个瓶颈则来自跨步骤累积的东西：上下文与轨迹管理。长时运行的智能体反复重新处理不断增长的历史、观测、工具输出与中间决策。ACON 通过针对下游智能体失败来优化压缩指南以解决这一问题，表明上下文压缩必须保留任务相关信息，而非简单地缩短文本 [277]。AgentDiet 从移除一侧补充了这一发现，证明长程轨迹包含可在推理时被剪除而不牺牲任务性能的冗余与过期信息 [425]。更近期的框架超越此类被动压缩，转向主动的上下文治理。ARC 将上下文视作一个动态的推理状态，可在出现"上下文腐烂"（context rot）时被监控与修订，而 Context as a Tool 则将上下文维护变成长程 SWE 智能体内部一个可调用的动作 [426, 427]。这些工作共同表明，高效的框架应将历史视作一个被管理的资源，其保真度、生命周期与可访问性由运行时治理，而非一份不可变的逐字记录。

在模型计算什么、看到什么上下文之外，效率还取决于工具与推理组件如何被编排。工具使用可通过将计算委派给外部系统来减轻模型负担，但结构糟糕的调用会引入额外调用、冗长的反馈轨迹与昂贵的重新规划，从而抵消委派的益处。近期关于分层工具编排的工作表明，将全局依赖指导与局部反思纠正分离，可以减少这种开销，避免在每次工具级失败后进行全轨迹重新规划 [428]。这三个维

度构成一个连贯的效率栈：路由控制使用哪个模型，上下文管理控制模型看到什么，轨迹缩减控制哪些历史得以持久，工具编排控制在协调外部动作上花费多少推理。因此，未来的框架应将计算视作一项有预算的运行时资源，在不确定性或任务风险足以证明其合理性的地方分配容量，同时将常规子任务坍缩为更便宜的模型调用、压缩的状态、缓存的产物或结构化的工具执行。

6.2 安全

智能体系统中的安全正从输出对齐转向运行时能力治理。当一个智能体只产生文本时，安全可围绕有害响应或偏好对齐来界定。一旦框架允许模型浏览不可信内容、调用工具、写入外部系统、存储持久记忆或委派子任务，主要的风险面便从生成转移到"观测到动作"的循环之中。这一扩张的攻击面在框架越来越深的层次上显现。在工具输出层面，AgentDojo 表明返回的结果可能携带提示注入攻击，将智能体引向恶意动作 [429]。在工具元数据层面，MCPTox 证明 MCP 风格的生态系统甚至在工具被执行之前就引入了投毒风险 [430]。在记忆层面，AgentSys 论证当不可信观测在工作记忆中累积时，间接提示注入会变得持久，并提出分层记忆隔离作为防御 [32]。这一递进表明，框架安全不仅关乎模型生成什么，也关乎运行时允许哪些信息、工具与权限去影响未来的动作。

认识到威胁面横跨整个运行时，近期研究越来越多地将安全视作一个治理架构问题。在概念层面，OpenPort Protocol 将工具访问表述为一个治理接口，结合依赖授权的发现、范围化权限、稳定的失败语义、风险门控的执行、状态重验证与结构化审计事件 [431]。在形式层面，"AI 智能体的运行时治理"表明许多安全策略是路径依赖的，即一个动作是否可接受取决于智能体身份、部分执行轨迹与先前的数据访问，而非孤立地取决于下一个动作 [432]。在实施层面，AgentSpec 提供了一种领域专用语言，用于在智能体动作上指定触发器、谓词与干预，而 AARM 则提出一个动作拦截层，累积会话上下文、检查策略对齐，并记录防篡改的回执 [433, 434]。SafeHarness 在完整的智能体生命周期内整合这些关切，结合对抗式上下文过滤、因果验证、特权分离的工具控制，以及带自适应降级的回滚 [435]。这些工作共同标志着一种转变：从提示级的保护措施转向框架内部生命周期感知的居间管控。

然而，这些治理机制大多处理单个动作或单层威胁。核心的开放问题是长时域上的组合式治理，其中不安全行为并非源自任何单个不被许可的步骤，而是源自一系列各自被许可步骤的组合：在发送邮件前读取敏感数据、在元数据被投毒后调用一个良性工具、在策略已变更的情况下检索陈旧记忆，或在不保留溯源的情况下委派部分上下文。应对此类组合式风险需要随状态一同传递的安全机制。工具输出、记忆条目、检索到的文档与计划中的动作都应携带可在下游使用前检查的溯源、信任与新鲜度元数据。高风险动作应支持审批门、状态重验证、在可能时回滚，以及在执行无法撤销时提供可审计的回执。这一表述厘清了框架级安全与模型级对齐的区别：对齐塑造行为倾向，而框架治理决定哪些能力被暴露、哪些信息得以持久、哪些动作可以执行，以及在执行前后必须产生什么证据。

6.3 持续学习

智能体系统中的持续学习不应等同于持续更新模型权重。智能体通过多个层次的适配来改进：模型可被微调，框架可更新其工具、规则与策略，上下文层可累积记忆、用户偏好与可复用技能。近期产品侧的讨论通过区分模型级、框架级与上下文级学习，使这种分层视角变得显式。一个有用的理论基础是 LLM 智能体的"外部化"视角，它观察到记忆将状态跨时间外部化，技能将程序性专长外部化，协议将交互结构外部化，而框架在运行时协调这些被外部化的组件。结合这些视角，持续学习便不再

那么关乎一个智能体是否拥有更大的记忆库，而更关乎框架如何维护那些使过去经验影响未来行为的外部结构 [436]。

如果持续学习主要作用于外部化状态，那么核心挑战便从记忆累积转向上下文生命周期管理。MemGPT 等早期系统引入了结构化的记忆层级，而非将上下文窗口当作只可追加的逐字记录 [148]。后续研究使记忆操作本身可学习：A-MEM 区分记忆构建与检索策略，而 Memory as Action 则将记忆编辑视作智能体策略的一部分，而非固定的启发式规则 [31, 437]。生产部署通过将不同的生命周期阶段付诸实施，进一步验证了这一趋势。例如，Anthropic 的上下文管理指南区分了压实、工具结果清除与记忆：压实压缩当前对话，工具结果清除移除可重新获取的输出，而记忆则跨会话持久化结构化的笔记 [438]。其底层洞见是不同类型的状态需要不同的生命周期处理：有些应留在活跃上下文中，有些应被摘要，有些因可重建而可被丢弃，还有些必须持久化到窗口之外以供未来会话使用。因此，持续学习成为一个对状态进行的、受治理的"写—管—读"循环，而非一个简单的检索问题。

当这一生命周期挑战从陈述性记忆延伸到程序性技能时，它变得更加尖锐。Agent Workflow Memory 表明智能体能够从先前轨迹中归纳出可复用的工作流，并检索它们以指导未来的动作序列 [146]。更近期的面向技能的系统将此类过程视作显式的包：技能可包含指令、脚本、元数据、测试与执行约束，并可被框架动态地发现、加载、共享或版本化。然而，技能比陈述性记忆更紧密地耦合于外部依赖，因此一旦成为持久对象，便获得了类似软件的维护风险。摘要可能过度压缩罕见的例外，旧的信念可能在环境变化后仍然存续，工具 API 可能使程序性例程失效，而低质量的记忆可能污染未来的行为。因此，未来工作应将持续学习作为受治理的持久化来研究：如何巩固经验、保留溯源、使陈旧状态过期、调和相互矛盾的更新、对技能进行版本化，并使持久结构与工具、环境和用户保持对齐。

6.4 状态与环境建模

智能体框架中的状态与环境建模主要并不关乎一个 LLM 是否包含潜在的常识知识。更直接的问题是运行时能否在长轨迹上维护一个紧凑、可更新、与动作相关的环境表征。对于实用智能体而言，外部世界仅通过观测（如网页变化、GUI 布局、仓库文件、工具响应、沙箱反馈与中间产物）被部分地暴露。由于这些观测不完整且短暂，将完整交互历史当作状态的代理既低效又脆弱：历史随冗余或过期信息增长，而模型的隐含信念可能悄然偏离实际环境。因此，框架需要显式机制来构建并更新一个支持规划、执行与恢复的信念状态，而非依赖原始观测的重放。

近期研究从两个互补方向应对这一需求，对应两种不同的规划要求。第一个方向聚焦于显式的信念维护，它回答"智能体当前处于何处"。WorldCoder 通过编写代码并与环境交互来学习可执行的环境模型，使规划能够在所学模型上而非在原始历史上进行 [439]。CoEx 将规划与一个从经验中更新的持久神经符号信念状态相耦合，而 PABU 表明进度感知的信念更新通过仅保留与决策相关的状态，可以胜过对完整历史的条件化 [440]。第二个方向建模动作条件化的转移，它回答"如果智能体行动会发生什么"。强化世界模型学习训练智能体预测动作后的下一状态，并使用仿真反馈与真实反馈之间的差异作为学习信号，使状态转移本身成为框架优化的一部分。网络、GUI 与移动环境中的相关工作类似地使用状态机或转移图来表示界面动态与动作后果。这两种能力共同使智能体能够同时推理其当前处境与候选动作的可能结果，而这正是有据可依规划的最低要求。

当学术研究追求所学状态模型时，已部署系统常通过结构化基础设施实现相同的功能。编码与计算机使用智能体通过仓库文件、终端输出、测试结果、执行日志与沙箱权限来维护环境状态。近期产品侧运行时使这一点变得显式：受控的沙箱环境定义智能体能检视、执行与修改什么，而渐进式披露则决定在每一步浮现哪些部分。Agent-World 通过论证稳健的智能体开发需要可扩展的环境合成、沙箱化与真实的状态转移，扩展了这一视角 [441]。无论是通过所学模型还是工程化基础设施实现，底层挑战都是相同的：维持智能体所相信的与实际为真的之间的一致性。因此，未来工作应将状态建模作为一个框架级的一致性问题来对待：如何调和模型先验、部分观测、工具反馈、记忆与变化的外部状态；如何检测陈旧或不确定的信念；以及如何跨网络、代码、GUI 与具身设定暴露可复用的状态转移接口。

6.5 具身框架

具身设定为框架工程提供了一个尤为苛刻的检验，因为状态是部分可观测的、执行充满噪声、反馈是实时的，且动作常常不可逆。PaLM-E 与 RT-2 等较早系统表明，多模态模型能够将视觉与语言知识连接到具身规划 [314, 315]，但这些系统大多将模型当作一个单体的"规划者到执行者"流水线。近期研究将这一流水线分解为显式的框架层。Gemini Robotics-ER 1.6 将高层具身推理（包括空间推理、任务规划与成功检测）与低层 VLA 执行及用户定义函数分离开来。Green-VLA 类似地将通用型机器人部署表述为一个分阶段的控制栈，在一个统一的动作接口之上设有任务规划、进度预测、分布外检测与基于 RL 的策略对齐 [442]。ACoT-VLA 与 ST4VLA 通过将动作空间推理与空间锚定作为 VLA 训练内的学习目标，强化了这种分解 [443, 444]。趋于一致的教训是，具身领域的进展更少依赖端到端的动作生成，而更多依赖高层语义推理与低层运动执行之间一条显式的运行时边界。

一旦确立了这条边界，锚定的状态管理便在语义层面成为框架的核心职责。与数字环境不同，物理世界不暴露完整或便于回滚的状态，因此框架必须从相机流、传感器读数与执行反馈中维护一个近似的信念状态。于是，三个运行时问题持续出现：当前观测是否足够可靠以据此规划，给定更新后的感知一个候选动作是否仍然有效，以及任务是否如预期般推进或已漂移出分布之外。近期系统通过具体机制应对这些问题：Gemini Robotics-ER 1.6 提供多视角成功检测与空间指向以进行进度评估；Green-VLA 增加了实时进度跟踪与 OOD 监控以检测漂移 [442]；而 RoboAgent 则将具身任务规划分解为由调度器居中调度的能力，并以独立的上下文进行可靠性隔离 [445]。共同的架构模式是，具身框架必须将感知、规划、信念更新与成功检测连接成一个连续的运行时循环，因为将感知当作一次性输入无法支撑多步的物理执行。

物理动作的不可逆性进一步要求超越软件智能体所需的执行前验证与恢复协议。在数字环境中，失败的动作往往可被重试或沙箱化；在物理环境中，单次不安全的抓取或碰撞可能损坏场景或机器人本身。EmbodiedBench 证实，即便是强大的多模态模型，在这些条件下也难以维持锚定与稳健的动作选择 [5]。因此，运行时不能仅依赖模型能力，而必须在将计划中的动作派发给控制器之前，验证它们与当前感知、物理约束与安全规则相兼容。AgentSpec 提供了一种有用的抽象，它将此类约束视作可执行的框架逻辑，而非非正式的提示 [446]。生产机器人工作流在规模上将同一原则付诸实施：NVIDIA 的"仿真到生产"技术栈与 Isaac Lab-Arena 要求在现实世界执行之前进行仿真评估、软件在环测试、数字孪生集群测试与安全护栏。当验证通过但执行仍然失败时，框架则必须支持重新观测、重新规划、控制器切换、退回安全状态，以及上报人类监督。从这个意义上说，具身框架与其说是一

个独立的应用领域，不如说是一个苛刻的试验台，用以检验为数字智能体开发的规划、状态维护、验证与恢复抽象能否在与物理现实接触时存活下来。

6.6 评测

针对智能体框架的评测正从度量孤立的任务成功，转向评估完整执行运行（run）的质量。近期基准已通过将智能体置于更长、更具交互性的设定中而超越了静态问答：BrowseComp 与 OSWorld 考验多步浏览与计算机使用轨迹，PaperBench 与 ReplicatorBench 评测开放式研究工作流，而 BeyondSWE 则瞄准那些只有在跨仓库推理之后才显现失败的软件工程任务 [313, 105, 73, 422, 408]。这些基准延长了任务时域与难度，但它们仍主要允许基于最终输出的评测。更近期的基准更进一步，使环境本身变得动态，从而迫使评测考虑整条执行轨迹。Gaia2 在异步环境中评测智能体，其中状态独立于智能体动作而变化，而 Claw-Eval-Live 则将可刷新的工作流信号与可复现的快照分离，并使用执行轨迹、审计日志与工作空间产物来给运行评分 [66, 447]。一旦环境可以在执行中途变化，一个正确的最终答案便不再保证一个稳妥的过程，从而使智能体的运行本身成为必要的度量单位。

然而，将运行作为评测对象，立即引出一个问题：该运行内部的什么应被度量。最终成功率掩盖了一个智能体究竟是遵循了一个可靠的程序，还是凭借偶然、过度重试或未被观察到的副作用到达了一个正确的终态。这一归因问题催生了第二项评测要求：过程可见性。现有基准已开始暴露更丰富的中间信号：ResearchRubrics 通过结构化评分标准评测长篇研究，OSWorld-MCP 使工具调用模式显式化，而 AgentLongBench 则通过环境 rollout 而非被动检索来评测长上下文智能体 [400, 70, 448]。这些努力指向一个更广泛的标准：基准应记录状态如何被更新、调用了哪些工具、产生了什么中间产物、失败如何被处理，以及记忆是否随时间变得陈旧。没有此类轨迹级证据，便仍难以将性能归因于模型、框架、工具或周围的执行环境。

过程可见性揭示一个智能体做了什么，但一次完整的评测还必须判断它所做的是否安全且负担得起。这催生了第三项要求：将治理、安全与成本作为一等指标来评测，而非次要的诊断。智能体失败越来越多地发生在轨迹内部，在那里它们对仅看输出的评测是不可见的：不安全的工具调用、被投毒的工具、被污染的记忆、通过内部消息的隐私泄露与缺失的审批步骤，可能从不会在最终响应中浮现。ATBench 通过异构工具池与延迟触发风险使这一轨迹级安全问题显式化，而 AgentLeak 则证明仅看输出的审计会遗漏通过智能体间消息与共享记忆的隐私泄露 [449, 450]。成本表现出一个平行的评测缺口。General AgentBench 表明测试时扩展受限于上下文上限与验证缺口，这意味着通过率应与 token 用量、延迟、工具调用次数与验证预算一并解读 [451]。因此，未来的框架基准应将路径质量、状态质量、安全质量与成本质量作为一个整合的画像来报告，以区分那些仅仅解决基准实例的系统与那些可靠、可审计、可部署的框架。

7 结论

本文从框架工程的视角对智能体系统进行了系统综述。从"实际能力既取决于系统脚手架，也同样取决于底层模型"这一观察出发，我们将框架形式化为掌管执行流程、工具调用、记忆管理、上下文构建与安全约束的编排层，并将框架工程表述为对脚手架及其所支撑模型的联合优化。在这一框架下，我们考察了智能体工作流、记忆系统、技能库与多智能体编排如何共同决定系统级性能，并回顾了横跨上下文工程与智能体训练的优化策略，以及覆盖软件工程、深度研究、工具使用、计算机使用与科学发现的评测基准。

这一视角将分析的焦点从智能体能完成什么，转向周围基础设施如何使可靠的完成成为可能。我们的分析揭示出，已部署系统中许多实际失败——包括长程执行期间的不稳定、低效的工具利用、上下文退化与次优的多智能体协调——往往源自框架设计的不足，而非仅仅源自模型级的缺陷。这些失败反映出 LLM 的单轮生成式接口与现实世界问题求解所固有的有状态、迭代特性之间的结构性失配，强化了本文的核心论点：模型与框架之间的关系本质上是协同增效的——对其中一方的改进会释放出另一方的潜在能力。

展望未来，若干方向值得探究：提升长程设定下框架执行的可扩展性与稳健性，发展有原则的脚手架–模型联合优化方法，将智能体框架推广到物理与具身领域，以及建立能够将框架贡献与模型能力区分开来的评测协议。我们希望本文能作为一份实用参考，助力通过有原则的框架工程构建更可靠、更可扩展、更可控的智能体系统。

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